

# Evapotranspiration measured via an eddy covariance network

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- I. Define Flux, ET
- II. Brief Primer on Eddy Covariance and how we use it to measure ET
- III. Networks of EC towers measuring ET across the globe
- IV. Evapotranspiration across the New Mexico Elevation Gradient



# Hydrologic Equation

- Inflow = outflow +/- Changes in storage
- Equation is simple statement of *mass conservation*

# Hydrologic inputs into area

- Precipitation
- Surface water inflow (streamflow + overland flow)
- Ground water inflow from outside area
- Artificial import (pipes + canals)

# Hydrologic outputs into area

- Evapotranspiration (Evaporation and Transpiration)
- Runoff of surface water
- Groundwater outflow
- Artificial export of water through pipes and canals



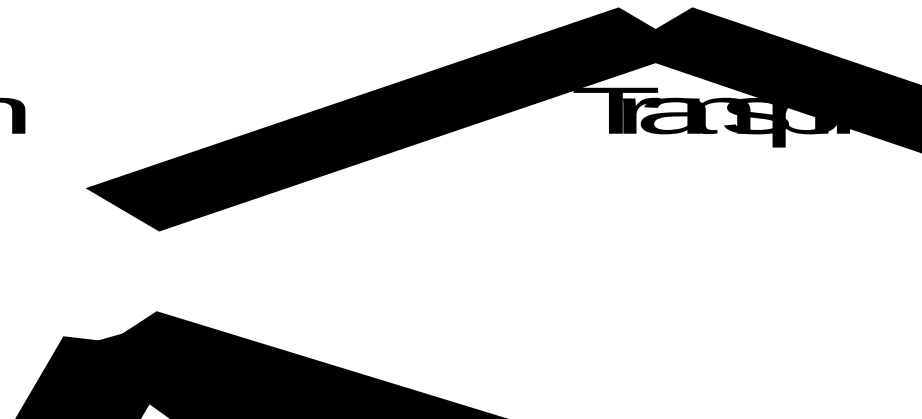


# EVAPOTRANSPIRATION

Evapotranspiration

Evaporation

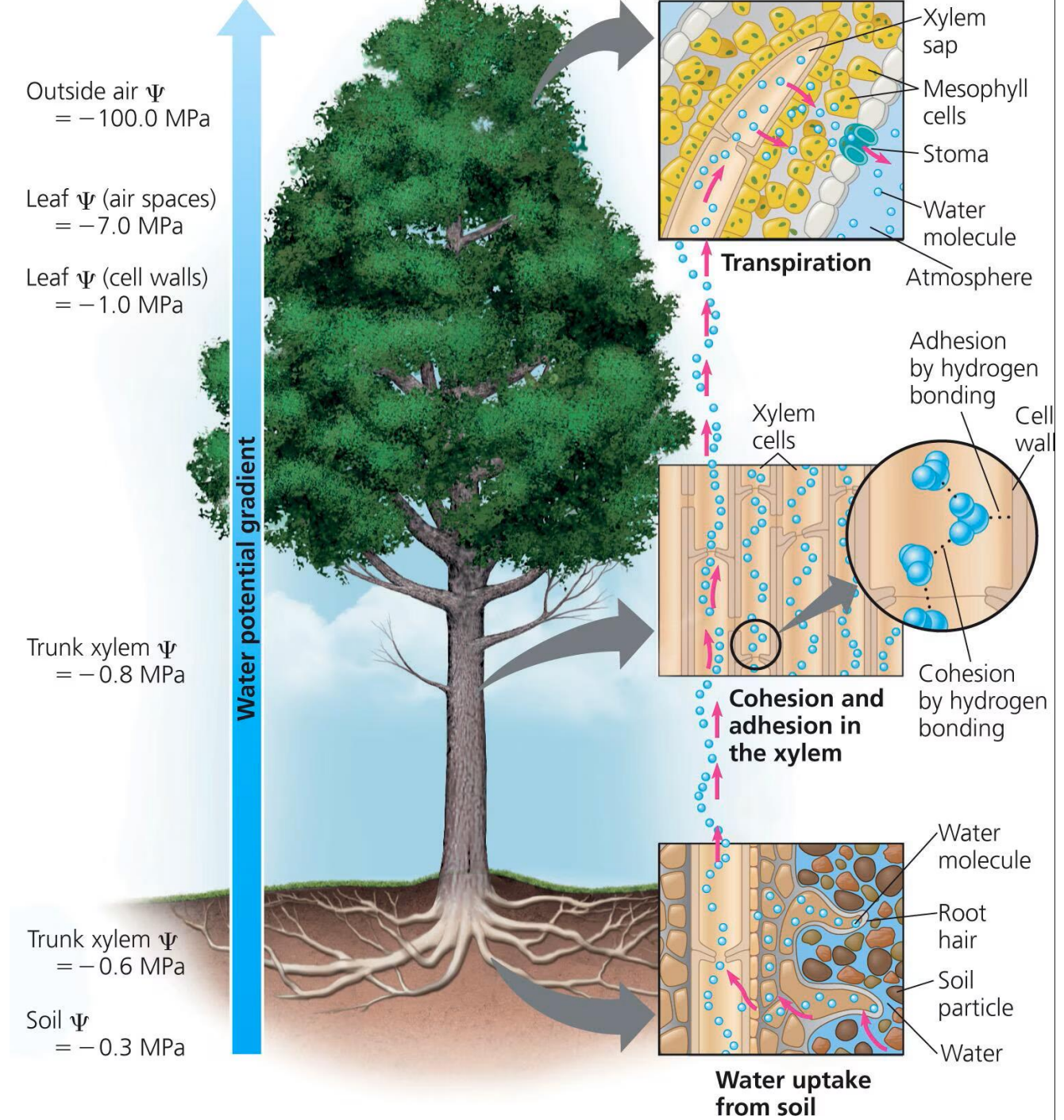
Transpiration



# What is transpiration and how does it work?

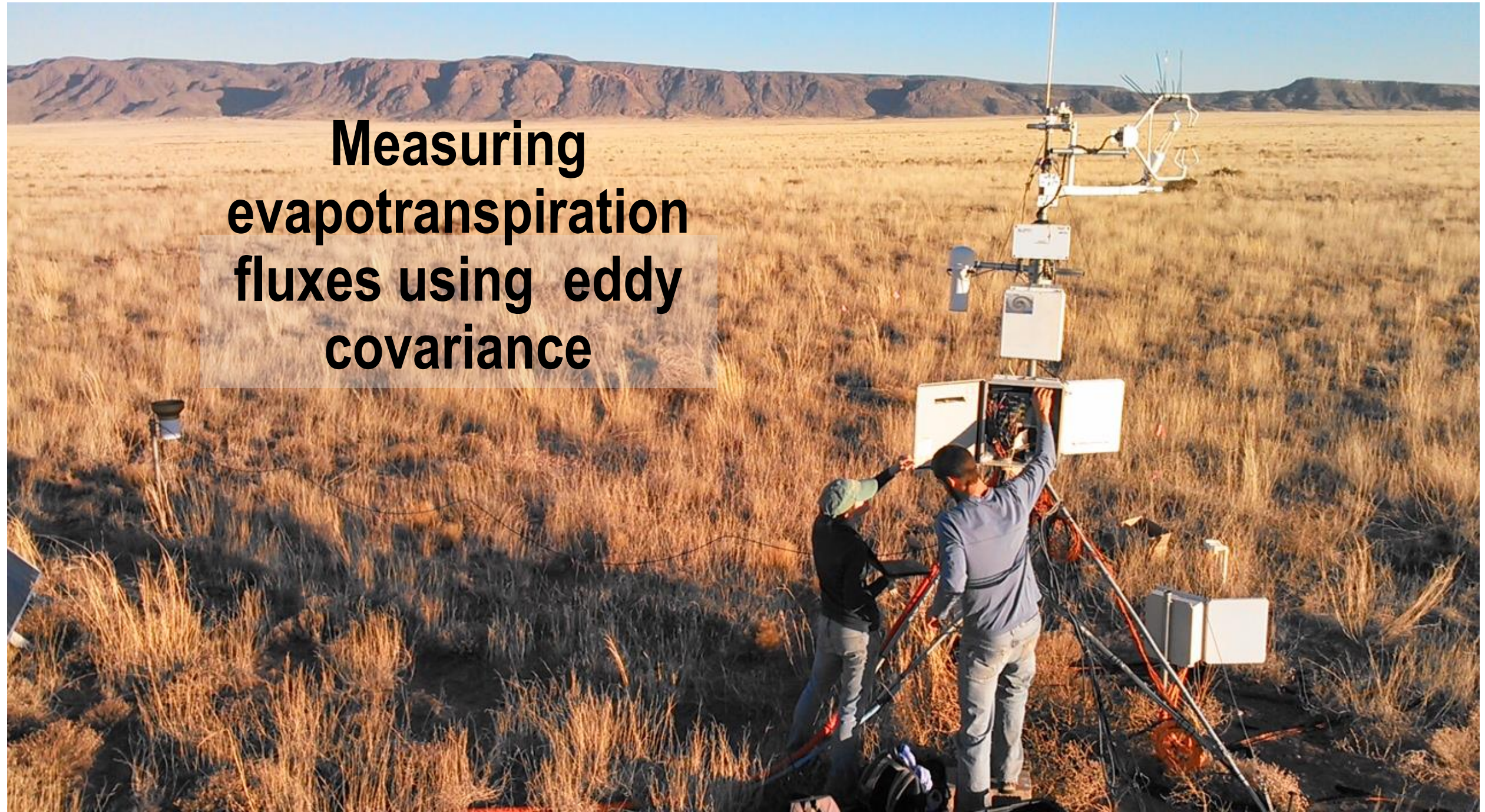
## Factors that regulate T:

- Solar radiation
- Net radiation
- Wind speed
- Temperature/Relative humidity
- Soil water availability
- Plant functional type
- Plant size/leaf area
- Plant density
- Community composition





**Measuring  
evapotranspiration  
fluxes using eddy  
covariance**





# What is a flux?

- Transport of stuff (scalar) expressed with respect to unit area and unit time
- Has a direction and sign
- Focus on surface flux (exchange between surface and atmosphere above it)
  - Carbon dioxide, Methane, VOC's, ozone, etc.....
  - Heat, Moisture, Momentum
- Carbon dioxide flux = “CO<sub>2</sub> flux” or Net Ecosystem Exchange (NEE)
  - $\mu\text{mol m}^{-2} \text{s}^{-1}$  or  $\text{g C m}^{-2} \text{day}^{-1}$
  - Positive (C added to atmosphere) or negative (C removed from atmosphere, added to ecosystem)
- Heat (Sensible Heat Flux)
  - $\text{Watts m}^{-2}$  (Watt = J/s)
  - Positive (Heat added to the atmosphere)
- Moisture (Latent Heat Flux = Evapotranspiration )
  - $\text{Watts m}^{-2}$  (Watt = J/s)
  - $\text{mm day}^{-1}$
  - Positive (moisture added to the atmosphere)

# Ecosystem-scale fluxes



- Site-level ecosystem process knowledge
  - Rates of carbon sequestration
  - Evapotranspiration rates
  - Rates of turbulent heat transfer to the atmosphere from the surface
  - Groundwater recharge
- Long-term trends, interannual variability

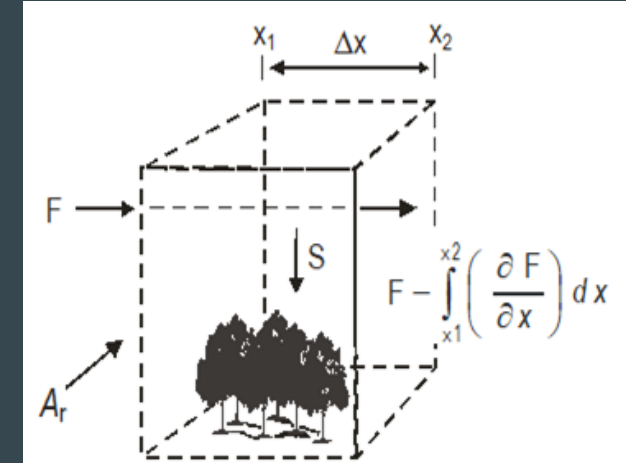
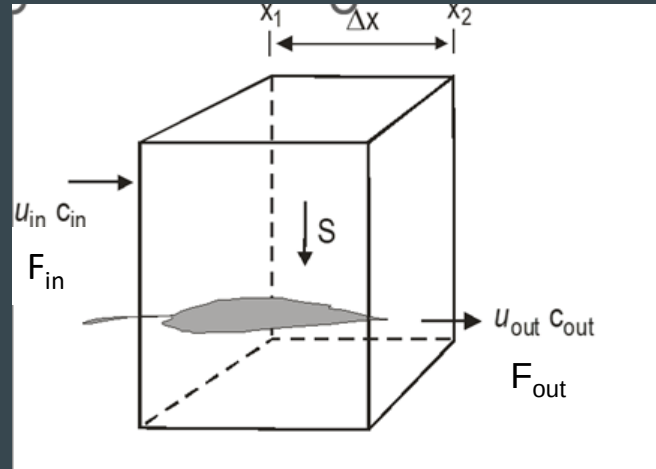


# Ecosystem-scale fluxes



- Site-level ecosystem process knowledge
  - Rates of carbon sequestration
  - Evapotranspiration rates
  - Rates of turbulent heat transfer to the atmosphere from the surface
  - Groundwater recharge
- Long-term trends, interannual variability - natural climate solutions
- Time-based information for inventory needs
- Disturbance (fires, insect outbreaks, fire)
- Hot moments
- Support management decision making
- Validate global models
- Parameterize global models
- Ground-truth remote sensing products
- Test ecological principles

# Fluxes measured with Licor chamber vs eddy covariance

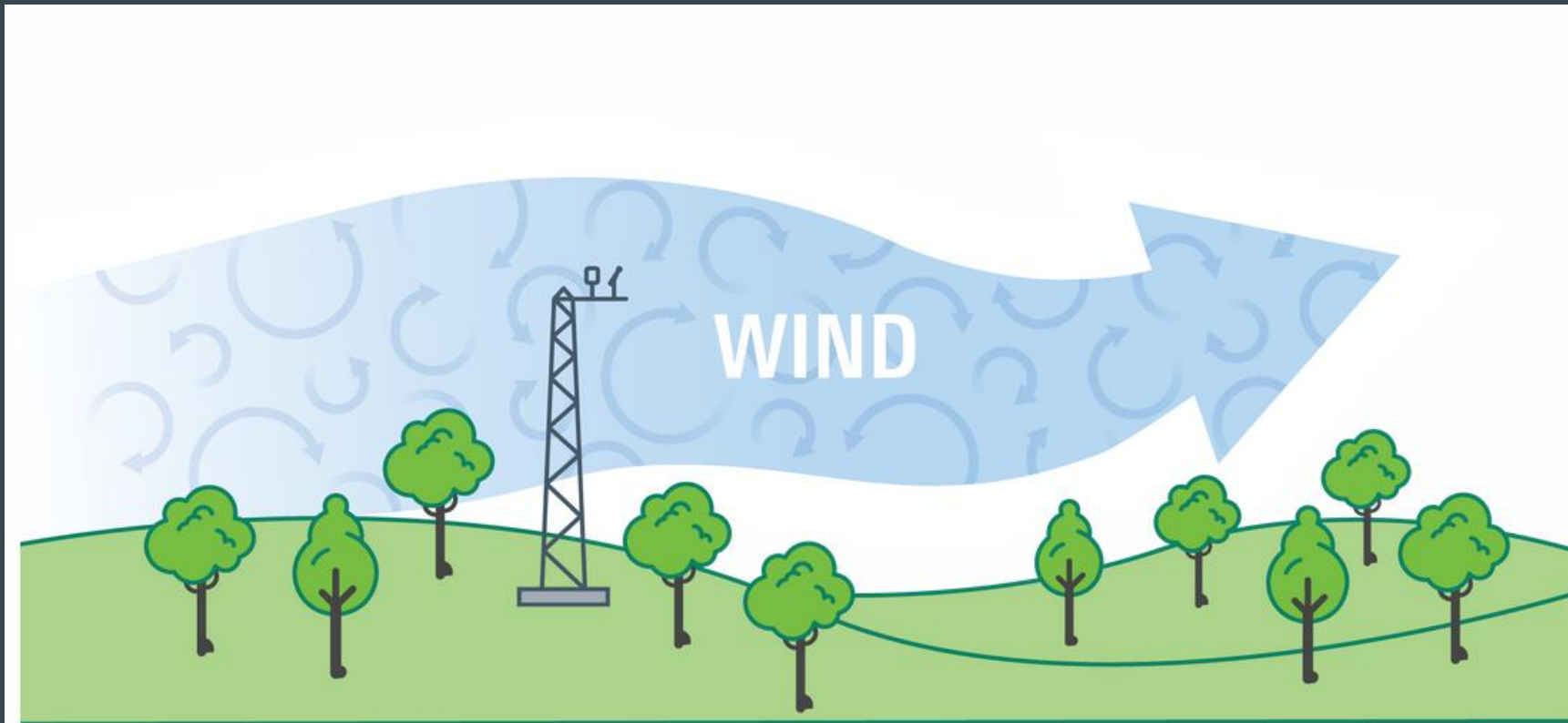


$$F_{in} = u_{in} c_{in} = \frac{\text{mol (air)}}{s} \times \frac{\text{mol (CO}_2\text{)}}{\text{mol (air)}} \times \frac{1}{\text{m}^2 \text{ (leaf area)}}$$

$$F_{out} = u_{out} c_{out} = \frac{\text{mol (air)}}{s} \times \frac{\text{mol (CO}_2\text{)}}{\text{mol (air)}} \times \frac{1}{\text{m}^2 \text{ (leaf area)}}$$

$$S = F_{in} - F_{out} = \frac{\text{mol CO}_2}{\text{m}^2 \text{ s}}$$





Burba, G., 2022. Eddy Covariance Method for Scientific, Regulatory, and Commercial Applications. LI-COR Biosciences, Lincoln, NE, USA; pp. 702

- Shear Stress
- Roughness
- Friction

The wind works within a 3-D coordinate system, so the flux vectors are not as neatly sorted as they are with a leaf chamber

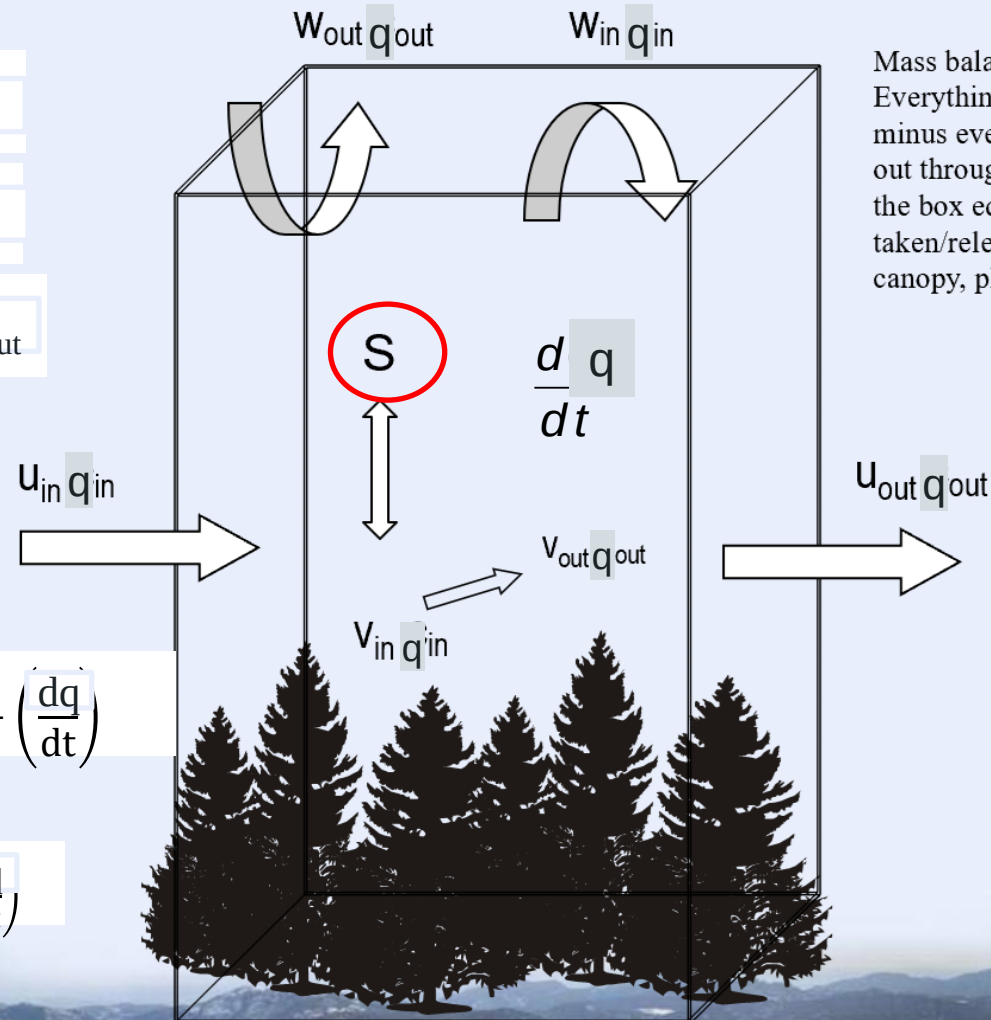
$$\Delta F_x = u_{in} q_{in} - u_{out} q_{out}$$

$$\Delta F_y = v_{in} q_{in} - v_{out} q_{out}$$

$$\Delta F_z = w_{in} q_{in} - w_{out} q_{out}$$

$$\frac{\Delta F_x}{\Delta x} + \frac{\Delta F_y}{\Delta y} + \frac{\Delta F_z}{\Delta z} = S + \left( \frac{dq}{dt} \right)$$

$$\frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z} = S + \left( \frac{dq}{dt} \right)$$



Can't forget storage (S)!

Mass balance:  
Everything that goes in  
minus everything that goes  
out through the surfaces of  
the box equals what is  
taken/released by the  
canopy, plus storage

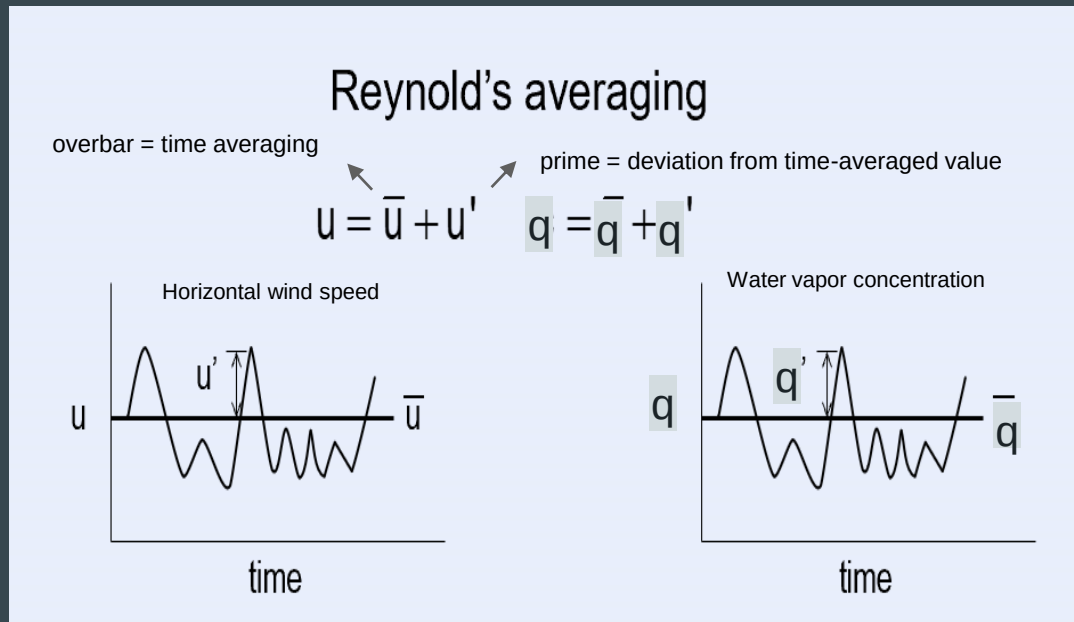


# Flows in the atmosphere and turbulence

At any instant in time, the speed recorded by an anemometer or h2o concentration recorded by an IRGA reflects a mean component and a turbulent component

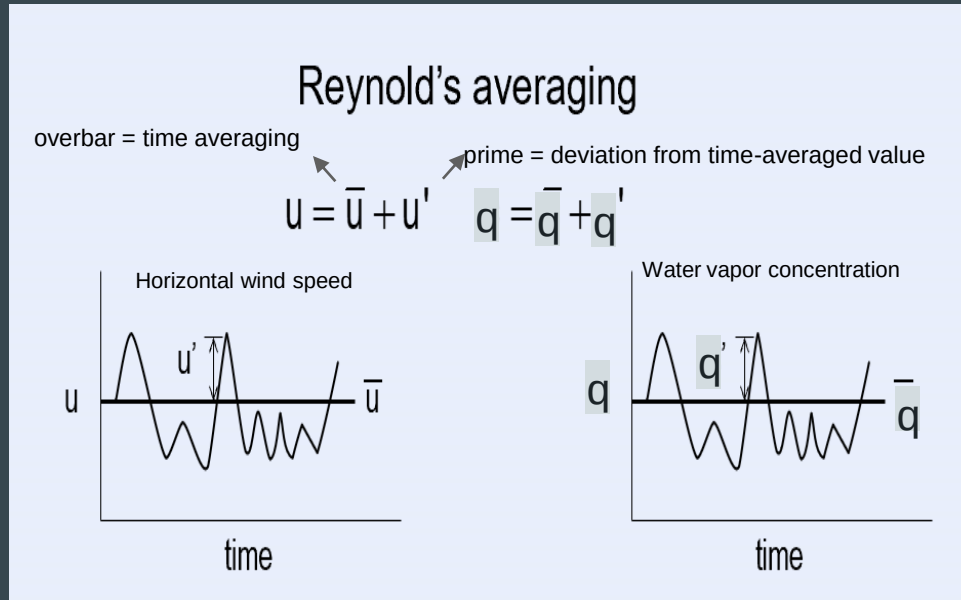


Sonic anemometer  
measures  
 $u'$ ,  $v'$ ,  $w'$  (wind speed)  
 $T'$  (temperature)



Gas analyzer  
measures:  
 $c'$  (CO<sub>2</sub>)  
 $q'$  (water vapor)

Statistical covariance between  $u$  and  $q$  represents the degree to which horizontal wind is moving water vapor concentration over a specific



Eddy flux = turbulent flux

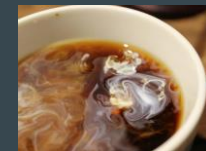
Time averaged multiplied product of 2 variances around means tells us to what extent the turbulent component of wind is moving the turbulent concentration of water vapor

we want the flux averaged over time  $\overline{uq}$

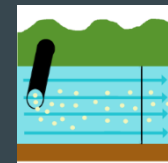
$$\overline{uq} = \overline{(\bar{u} + u')(\bar{q} + q')} = \overline{\bar{u}\bar{q}} + \cancel{\overline{\bar{u}q'}} + \cancel{\overline{\bar{q}u'}} + \overline{q'u'} = \bar{u}\bar{q} + \overline{q'u'}$$

$\bar{q}' = 0 \quad \bar{u}' = 0$

turbulence

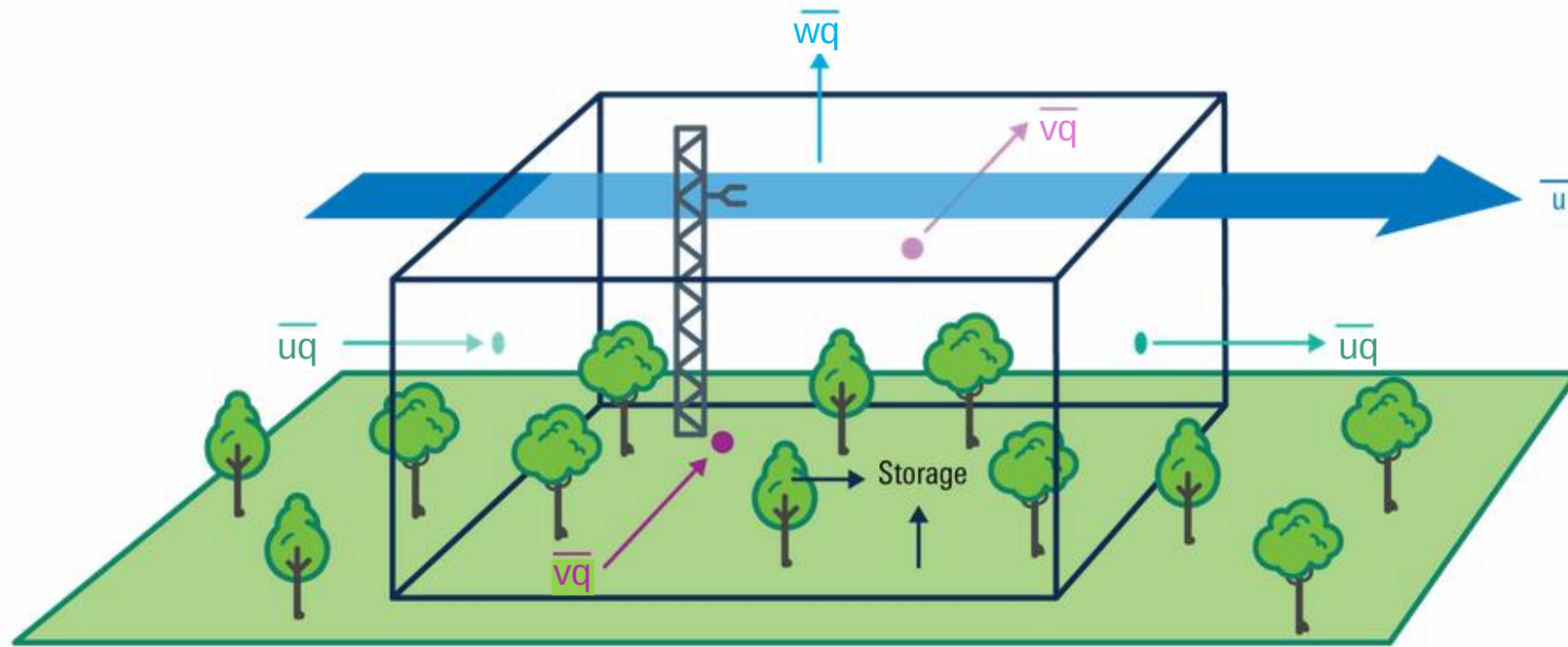


advection



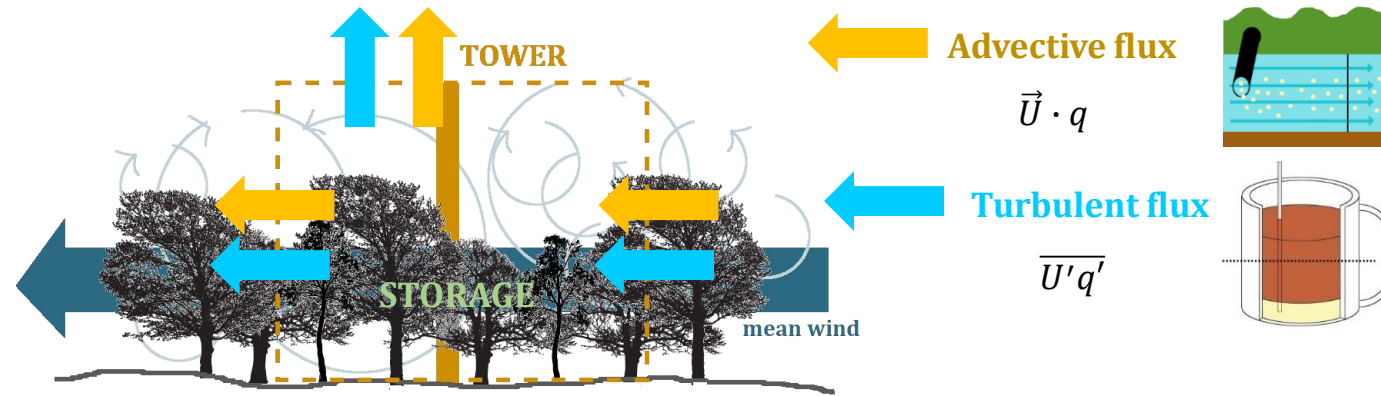
Time averaged multiplied product of 2 means tells us to what extent the mean wind flow is moving the mean concentration of water vapor





Burba, G., 2022. Eddy Covariance Method for Scientific, Regulatory, and Commercial Applications. LI-COR Biosciences, Lincoln, NE, USA; pp. 702

## Fundamental Assumptions and limitations



Evapotranspiration (ET)

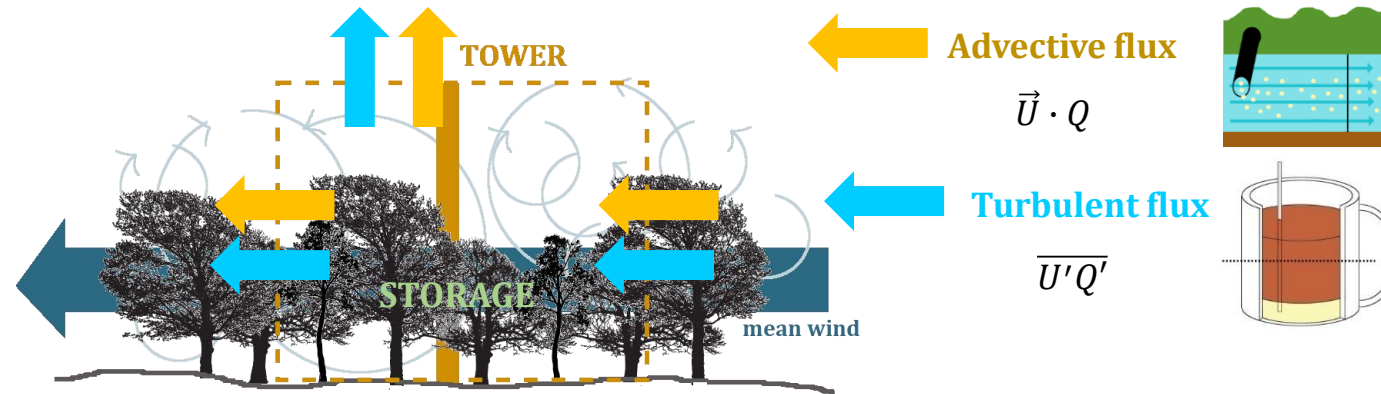
$$ET = \int_0^z \frac{\partial \bar{u}\bar{q}}{\partial x} dz + \int_0^z \frac{\partial \bar{v}\bar{q}}{\partial y} dz + \int_0^z \frac{\partial \bar{w}\bar{q}}{\partial z} dz + \int_0^z \frac{\partial \overline{u'q'}}{\partial x} dz + \int_0^z \frac{\partial \overline{v'q'}}{\partial y} dz + \int_0^z \frac{\partial \overline{w'q'}}{\partial z} dz + \int_0^z \frac{\partial \bar{q}}{\partial t} dz$$

Advective fluxes in 3 directions

Turbulent fluxes in 3 directions

Storage

# Fundamental Assumptions and limitations



## Fundamental Assumption #1: Sufficient Turbulence!

*Why: Eddy covariance measures turbulent fluxes, not advective fluxes!*

$$\text{ET} = \int_0^z \frac{\partial \bar{u}\bar{q}}{\partial x} dz + \int_0^z \frac{\partial \bar{v}\bar{q}}{\partial y} dz + \int_0^z \frac{\partial \bar{w}\bar{q}}{\partial z} dz + \int_0^z \frac{\partial \overline{u'q'}}{\partial x} dz + \int_0^z \frac{\partial \overline{v'q'}}{\partial y} dz + \int_0^z \frac{\partial \overline{w'q'}}{\partial z} dz + \int_0^z \frac{\partial \bar{q}}{\partial t} dz$$

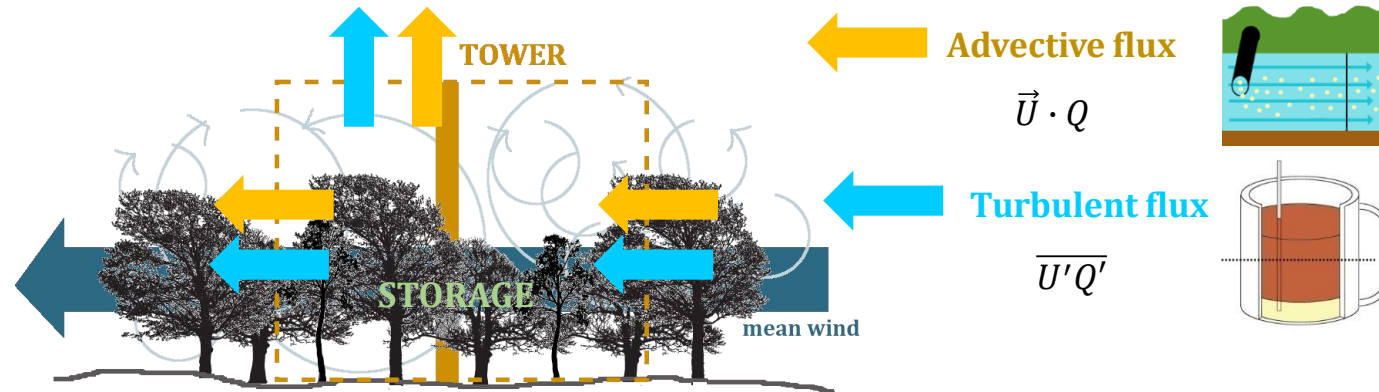
Advective fluxes in 3 directions

Turbulent fluxes in 3 directions

Storage



## Fundamental Assumptions and limitations



### Fundamental Assumption #2: Horizontal homogeneity!

*The horizontal fluxes coming in are statistically the same as the horizontal fluxes coming out*

*So, we can ignore them!*

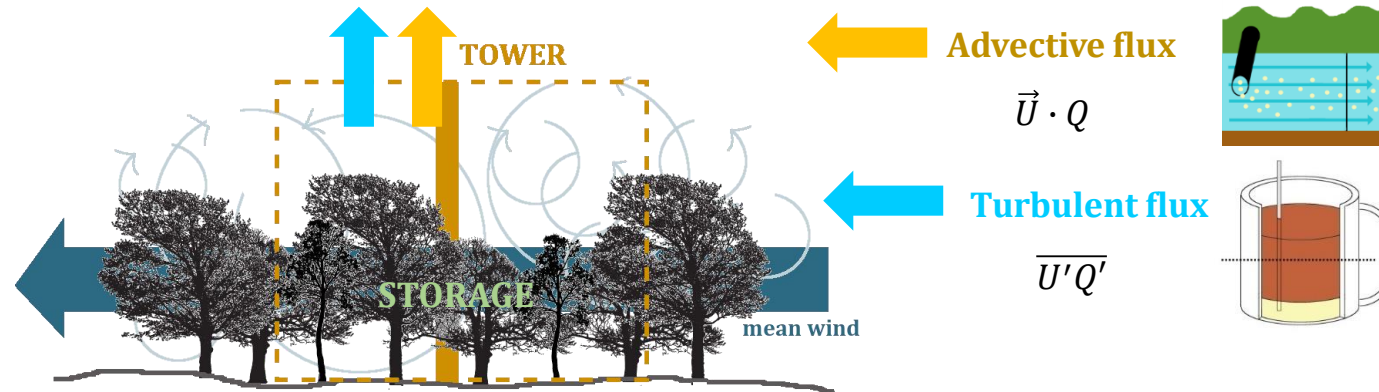
$$\text{ET} = \int_0^z \frac{\partial \overline{uq}}{\partial x} dz + \int_0^z \frac{\partial \overline{vq}}{\partial y} dz + \int_0^z \frac{\partial \overline{wq}}{\partial z} dz + \int_0^z \frac{\partial \overline{u'q'}}{\partial x} dz + \int_0^z \frac{\partial \overline{v'q'}}{\partial y} dz + \int_0^z \frac{\partial \overline{w'q'}}{\partial z} dz + \int_0^z \frac{\partial \overline{q}}{\partial t} dz$$

**Advection fluxes in 3 directions**

**Turbulent fluxes in 3 directions**

**Storage**

## Fundamental Assumptions and limitations



### Fundamental Assumption #3: no mean vertical wind speed

*In this case, vertical advection is zero...we can ignore it too!*

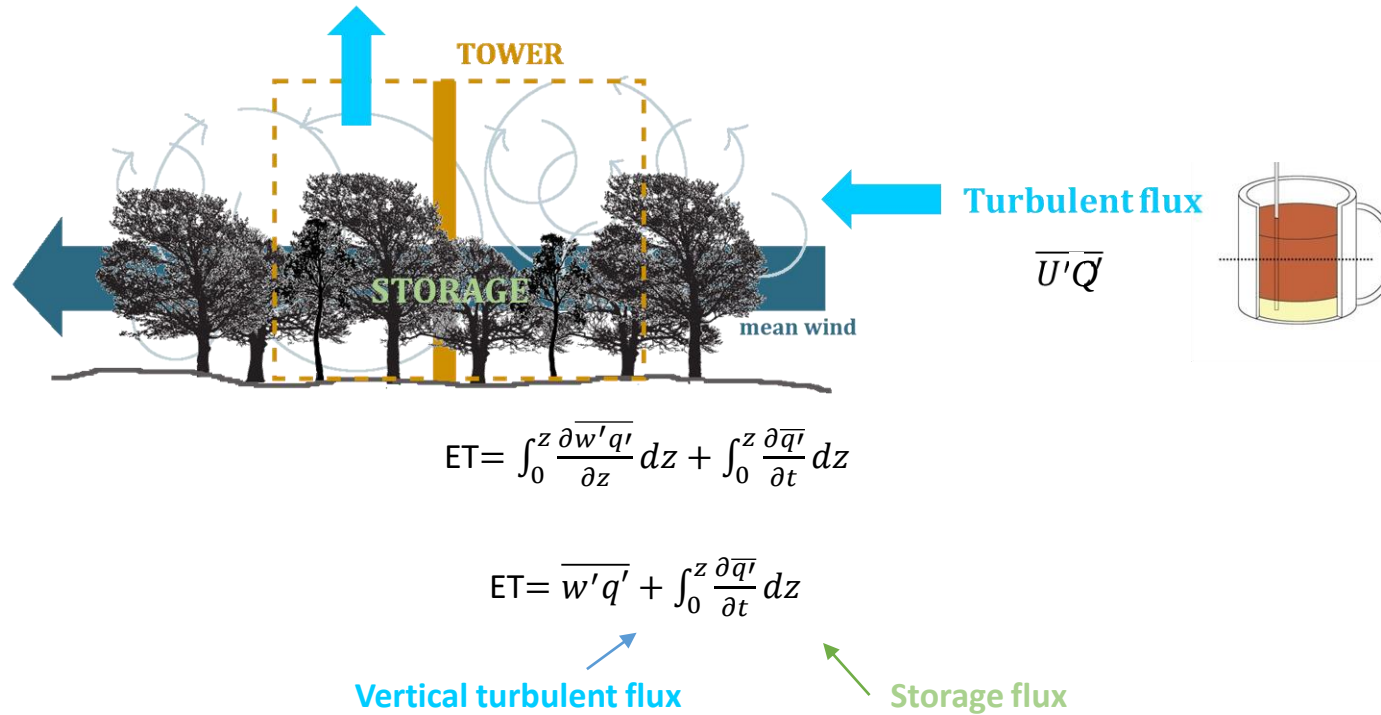
$$\text{ET} = \int_0^z \frac{\partial \overline{uq}}{\partial x} dz + \int_0^z \frac{\partial \overline{vq}}{\partial y} dz + \int_0^z \frac{\partial \overline{wq}}{\partial z} dz + \int_0^z \frac{\partial \overline{u'q'}}{\partial x} dz + \int_0^z \frac{\partial \overline{v'q'}}{\partial y} dz + \int_0^z \frac{\partial \overline{w'q'}}{\partial z} dz + \int_0^z \frac{\partial \overline{q}}{\partial t} dz$$

Advection fluxes in 3 directions

Turbulent fluxes in 3 directions

Storage

## Fundamental Assumptions and limitations



**Fundamental Assumption #4: Storage is negligible or measured with a profile system**



Okay to assume negligible storage flux here



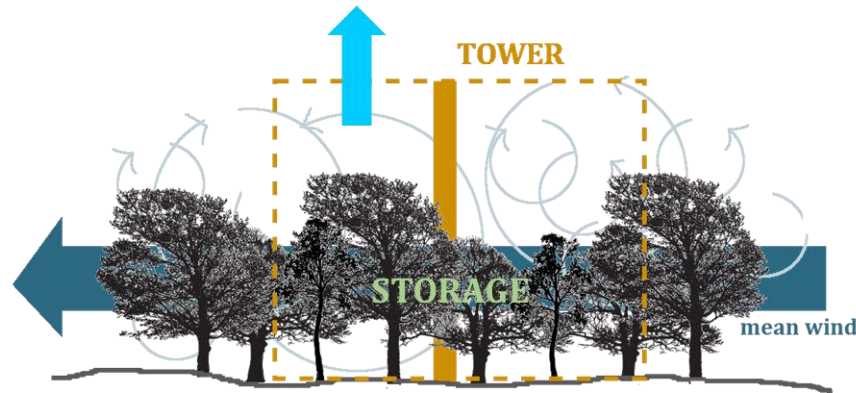
In forests, you should have a profile system



Every hour...find the change in concentration from one averaging period to the next at each level, then add them up (numerical integration)

$$\int_0^z \frac{\partial \bar{q}'}{\partial t} dz$$

## Fundamental Assumptions and limitations



### Evapotranspiration

$$ET = \overline{w'q'}$$

When all these assumptions are met, vertical ET is the thing we measure with eddy covariance

Sonic anemometer  
 $w'$  (wind speed)  
 $T'$  (temperature)



Gas analyzer :  
 $c'$  (CO2)  
 $q'$  (water vapor)

These are sufficient to calculate not only  $\overline{w'c'}$  (NEE, or carbon flux), but also  $\overline{w'q'}$  (water flux) and  $\overline{w'T'}$  (sensible heat flux)

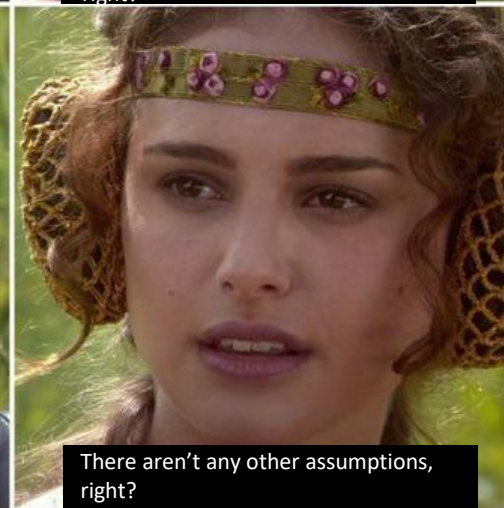
High frequency measurements (10x per second), capture fluxes generated by even the smallest eddies



That's how EC works



There aren't any other assumptions,  
right?



There aren't any other assumptions,  
right?

Instrument considerations (open vs closed path IRGA's)

Tower placement/height

Instrument placement

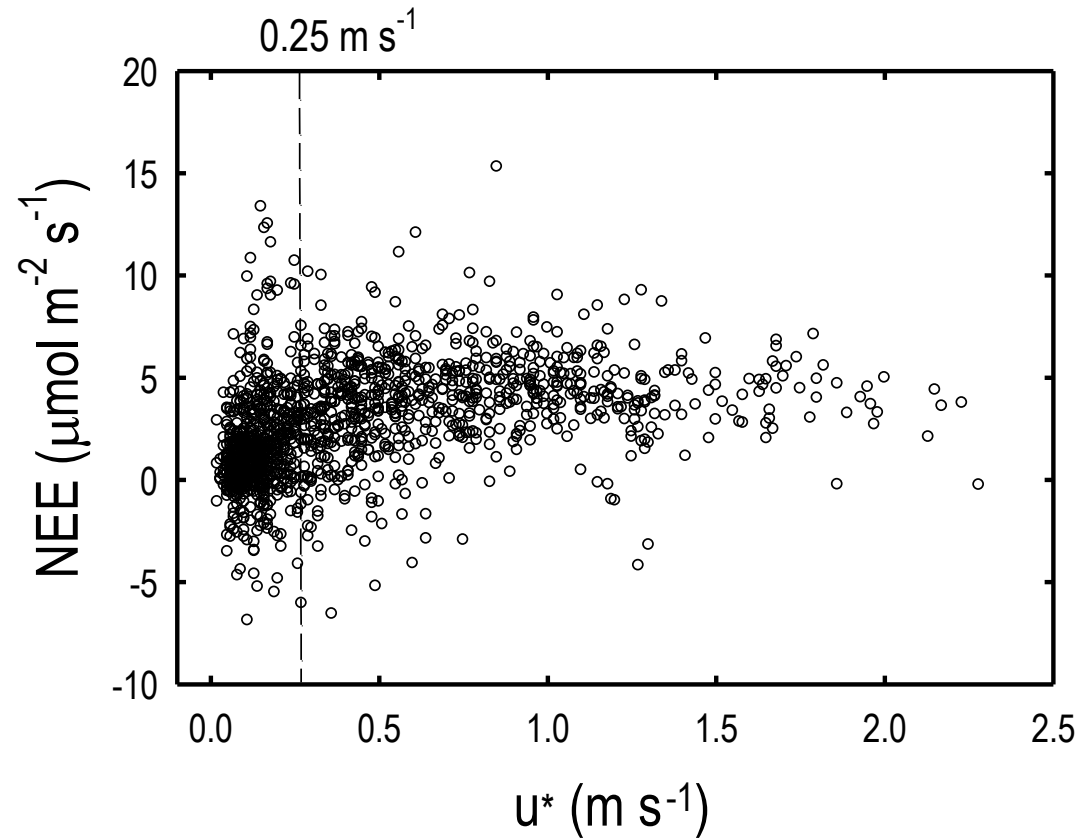
Energy balance closure



## Some practical considerations in applying eddy flux theory



The eddy flux measure is only reliable with high shear and adequate vertical turbulence ( $w'$ )



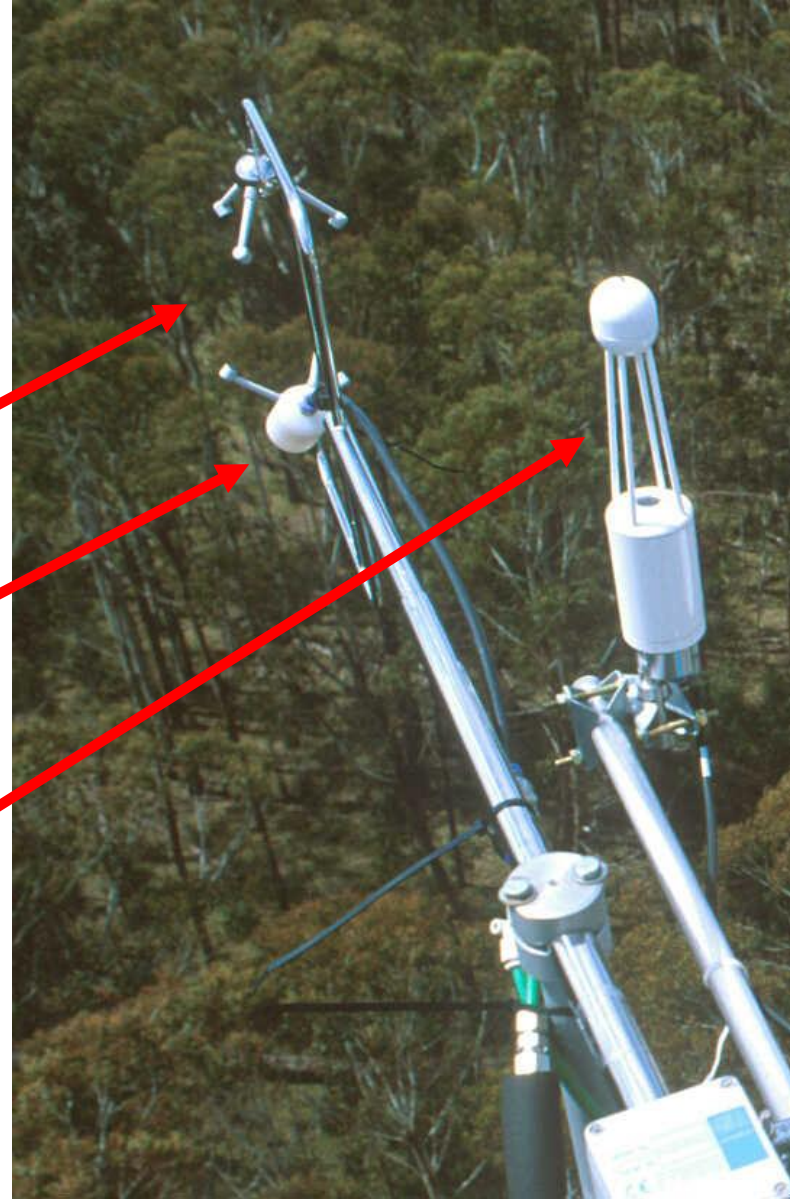
$$u^* = \sqrt{|\overline{(u'w')}|_0}$$

# Typical eddy flux instrumentation

Sonic anemometer

Air intake for  
closed-path CO<sub>2</sub> &  
H<sub>2</sub>O analyser

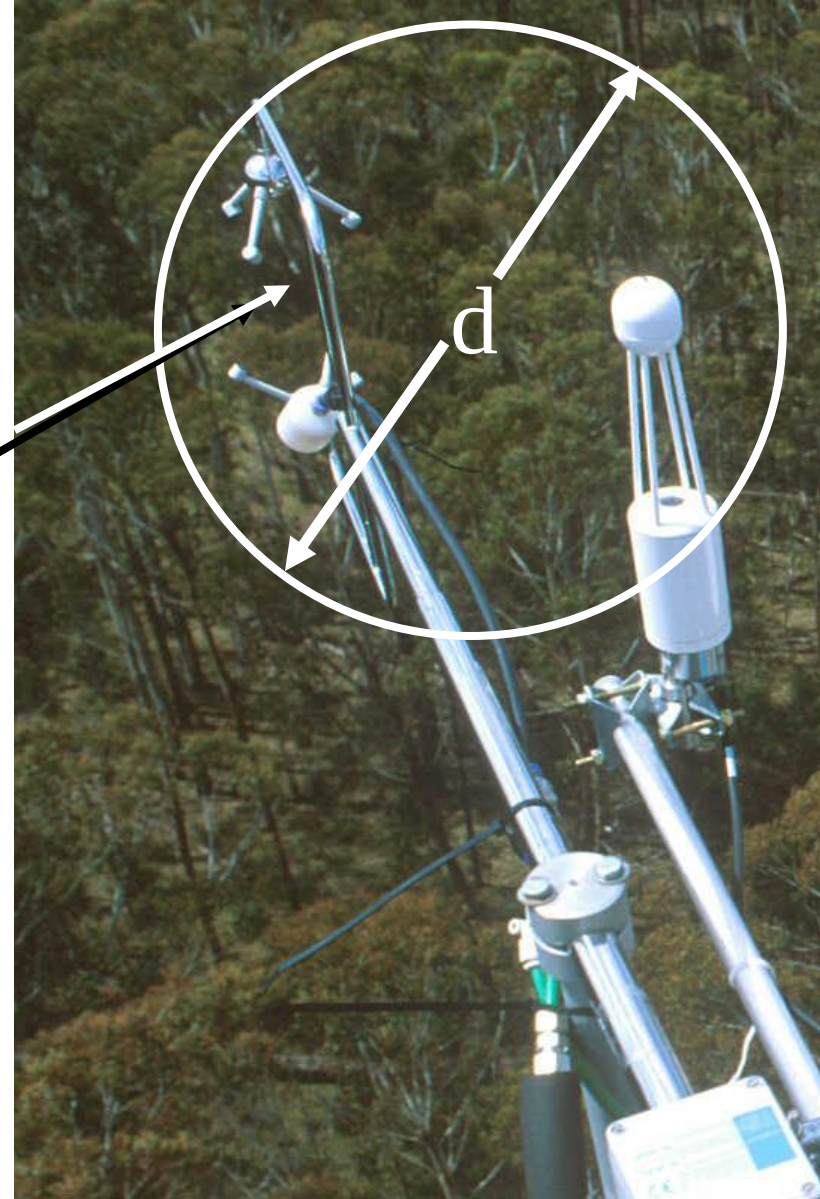
Open-path CO<sub>2</sub>  
& H<sub>2</sub>O analyser



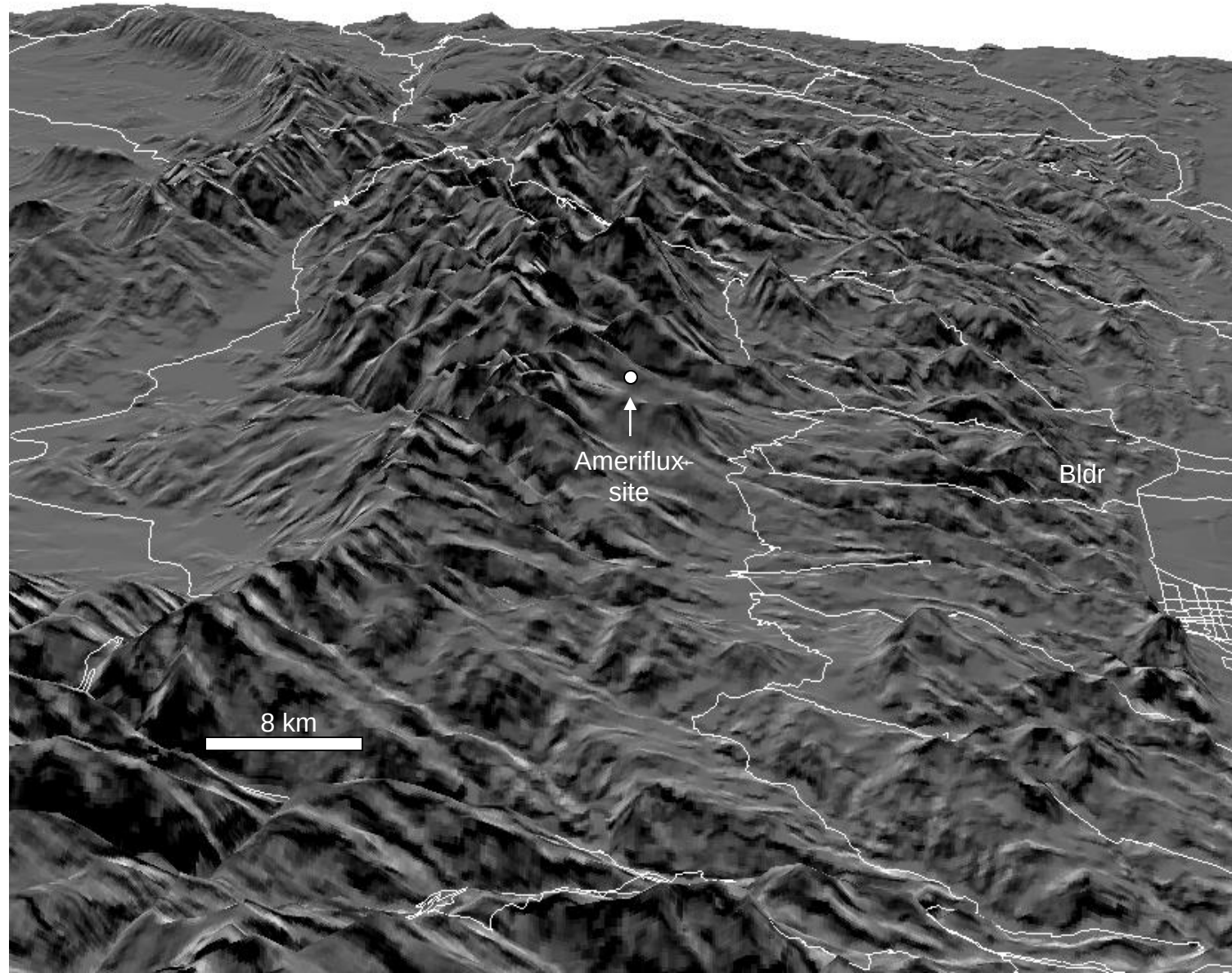


The instruments place physical constraints on our ability to measure the eddy flux

- loss of covariance
- shadowing of instruments
- line averaging



## The constraint imposed by complex terrain



The topographic complexity confounds approaches typically used to reduce the mathematical complexity

$$\text{NEE} = \int_0^z \frac{\partial \overline{u'c'}}{\partial x} dz + \int_0^z \frac{\partial \overline{v'c'}}{\partial y} dz + \int_0^z \frac{\partial \overline{w'c'}}{\partial z} dz + \int_0^z \frac{\partial \overline{u'c'}}{\partial x} dz + \int_0^z \frac{\partial \overline{v'c'}}{\partial y} dz + \int_0^z \frac{\partial \overline{w'c'}}{\partial z} dz + \int_0^z \frac{\partial \overline{c}}{\partial t} dz$$

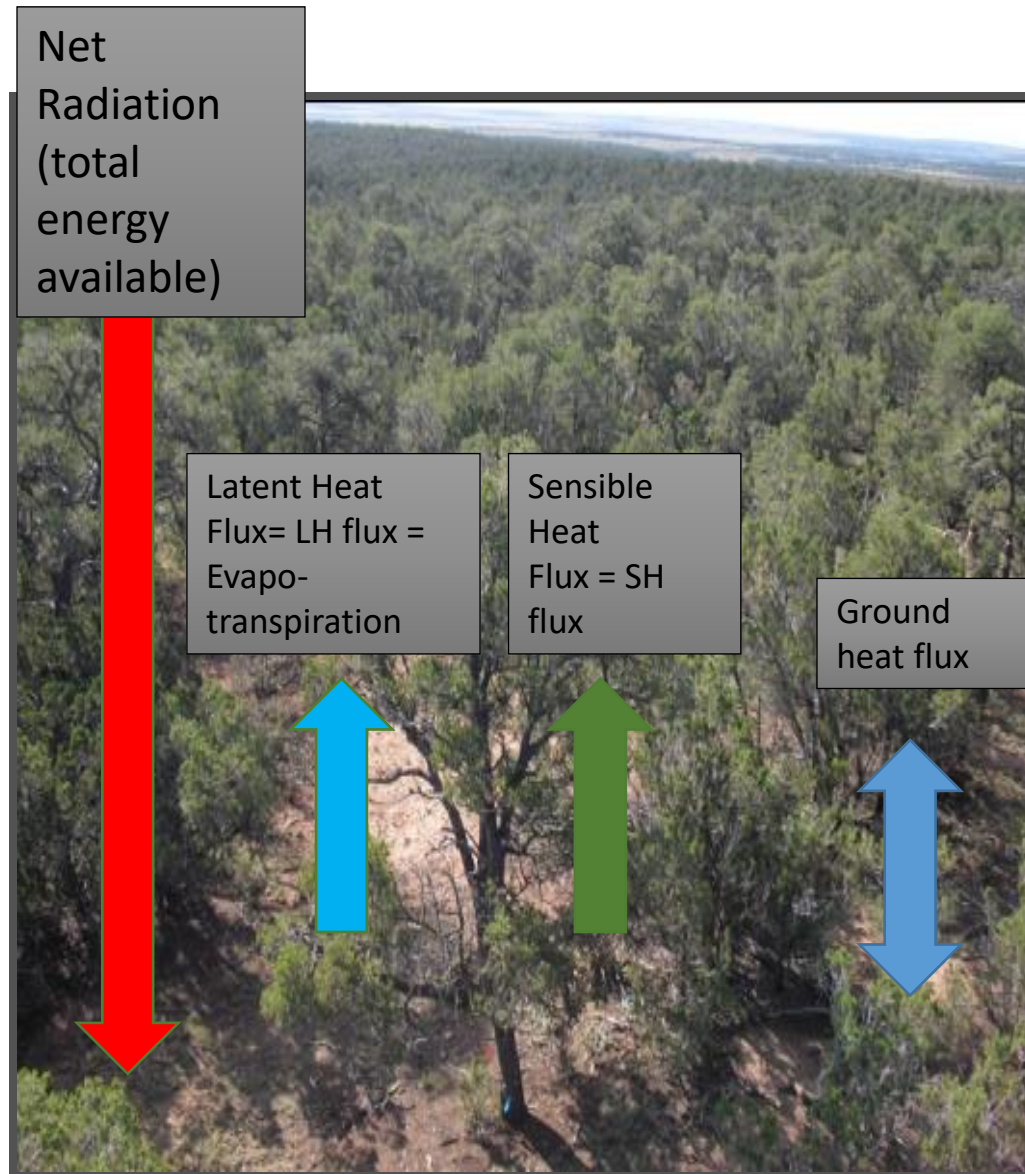
0      0      0      0      0

$\int_0^z \frac{\partial \overline{w'c'}}{\partial z} dz = (\overline{w'c'})_{z_{\text{ref}}}$

A general problem in atmospheric uncoupling



# Eddy covariance – ecosystem water and energy fluxes



$$\text{Energy IN} = \text{Energy OUT}$$

Net Radiation =  
LH flux + SH flux + ground heat flux

$\text{W m}^{-2} \text{ time}^{-1}$  (30 min, day, month, yr)

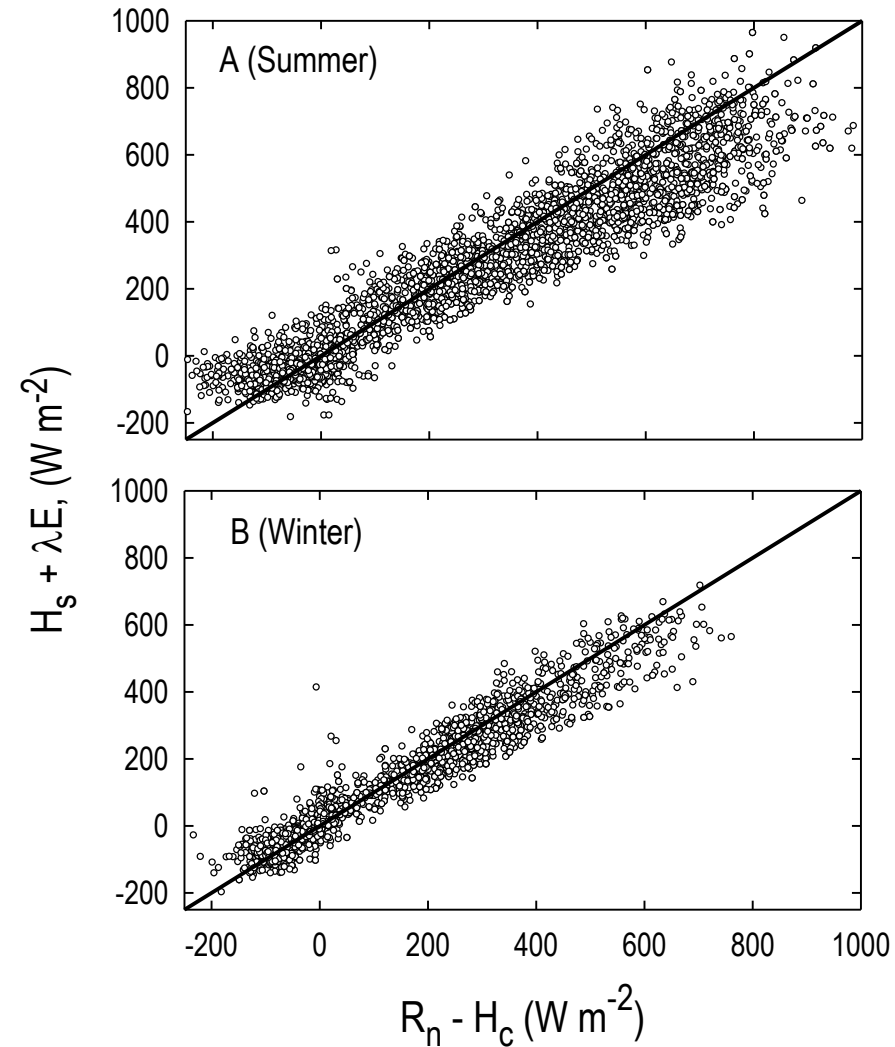
How much water is ecosystem using? How does climate influence this?  
How much water pumped into the atmosphere?  
How much energy used to heat atmosphere?  
How much water recharging groundwater?



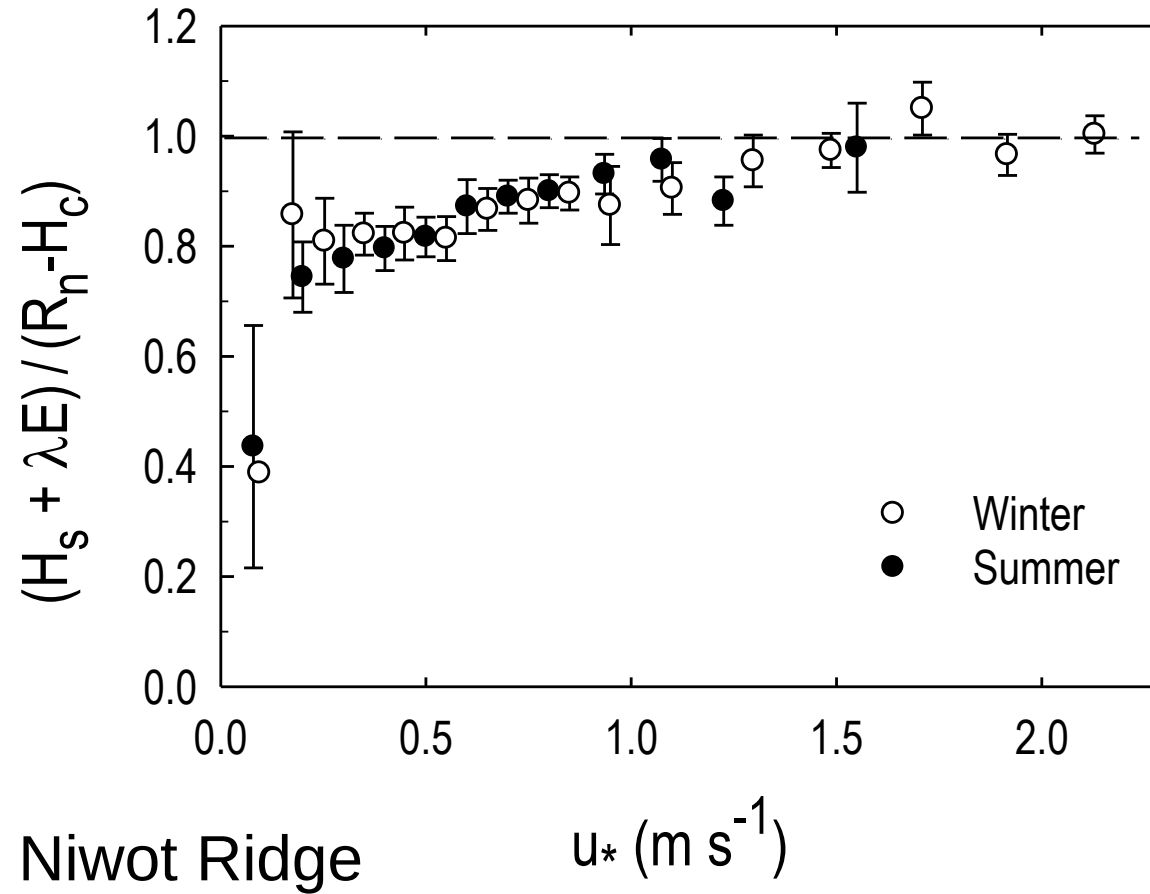
How do we know  
the fluxes are right?

The 'energy balance  
closure approach'

## Niwot Ridge



We get good closure with high shear and good turbulent mixing





## Review

## Energy balance closure at FLUXNET sites revisited

Matthias Mauder<sup>a,\*</sup>, Martin Jung<sup>b</sup>, Paul Stoy<sup>c</sup>, Jacob Nelson<sup>b</sup>, Luise Wanner<sup>a</sup><sup>a</sup> Institute of Hydrology and Meteorology, TUD Dresden University of Technology, Tharandt 01737, Germany<sup>b</sup> Department of Biogeochemical Integration, Max Planck Institute for Biogeochemistry, Jena 07745, Germany<sup>c</sup> University of Wisconsin-Madison, United States

## A B S T R A C T

The FLUXNET network with numerous eddy covariance stations distributed worldwide is an important backbone for the study of ecosystem-atmosphere interactions. In order to provide reliable data for a variety of related research fields all parts of the ecosystem-atmosphere interactions need to be fully captured. Energy balance closure can be an indicator that all fluxes are fully recorded. However, in an investigation of the FLUXNET data set over 20 years ago, a systematic imbalance of around 20 % was observed in the surface energy balance. By improving measurement instruments and arrangements as well as data post-processing, the imbalance was reduced to about 15 % within the following ten years. We show that the remaining imbalance has hardly changed to this day. In the meantime, it has become clear that the energy transport through mesoscale secondary circulations, which by definition cannot be captured with single-tower eddy covariance measurements, accounts for a large proportion of the remaining imbalance and leads to an underestimation of atmospheric energy fluxes. Storage changes, which have so far only been partially recorded, were also found to strongly contribute to the imbalance. In addition to recommendations for improving storage change measurements, we therefore present various energy balance closure approaches. These can be used to complement FLUXNET measurements by accounting for those flux contributions that cannot be captured by single-tower measurements or to parameterize the transport by secondary circulations in numerical weather and climate models. Another important finding in energy balance closure research is that secondary circulations contribute not only to energy transport but also to the transport of CO<sub>2</sub> and other substances, but more research is needed in this area. We conclude that research into energy balance closure problem has made great progress in recent years, which is crucial for investigating the role of ecosystems in the Earth system.



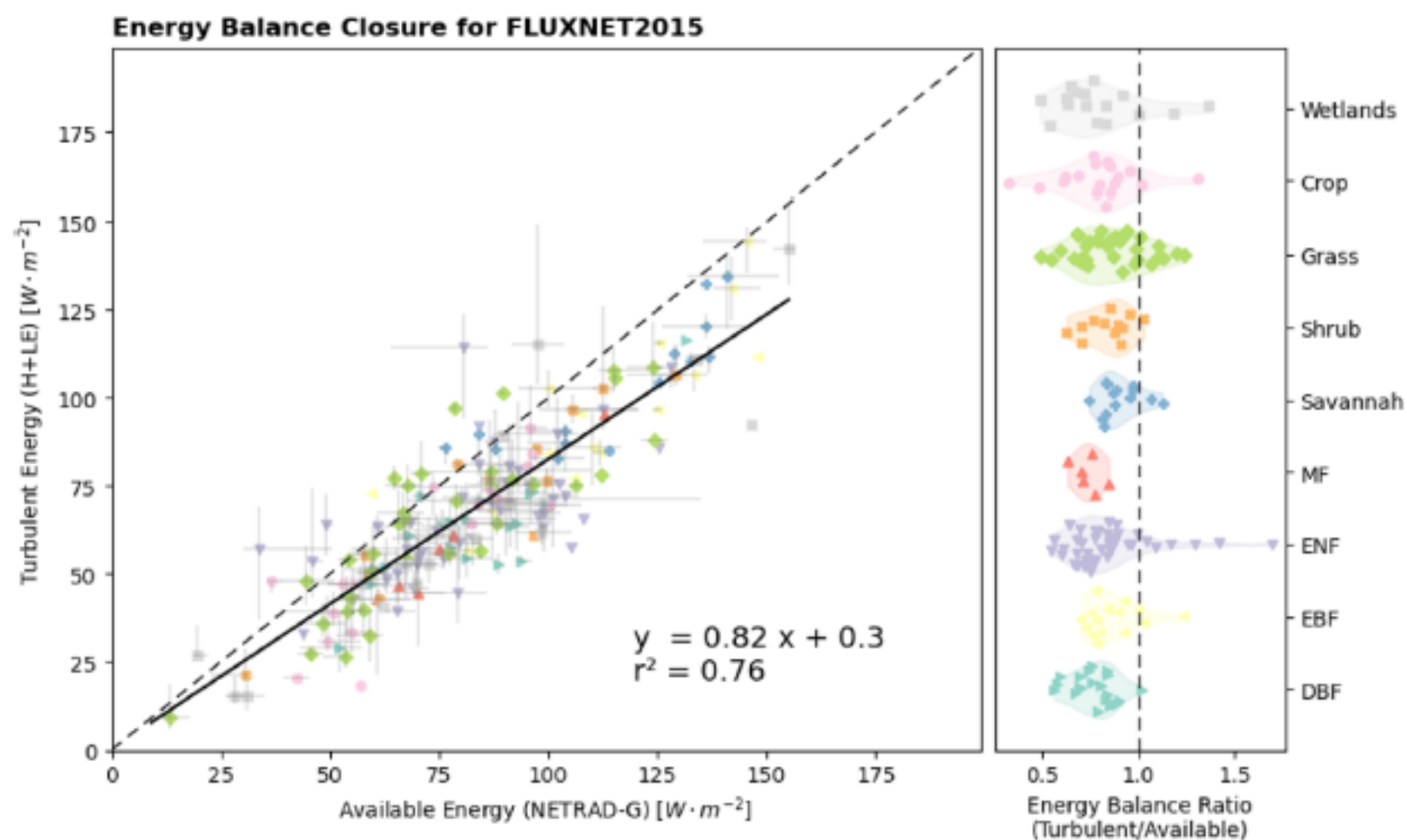


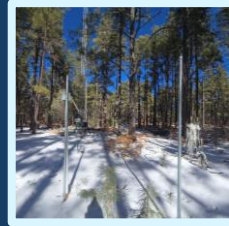
Fig. 1. Energy balance closure of the FLUXNET2015 dataset (Pastorello et al., 2020). Points represent the median across years of mean annual available energy (net radiation minus ground heat flux) to turbulent energy (sensible plus latent heat flux). Error bars represent the variability (5th to 95th percentiles) across years, with all years where at least 60 % of all fluxes were directly measured or gap-filled with high quality estimates (ONEFlux flags 0 or 1). In cases where ground heat flux was not measured, it was assumed to be net zero over the year.

# New Mexico Elevation Gradient

Litvak Lab, UNM  
2007 - present



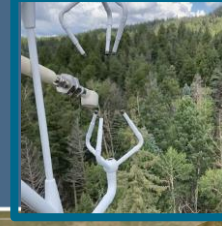
ponderosa pine



post-fire mixed  
conifer



mixed conifer



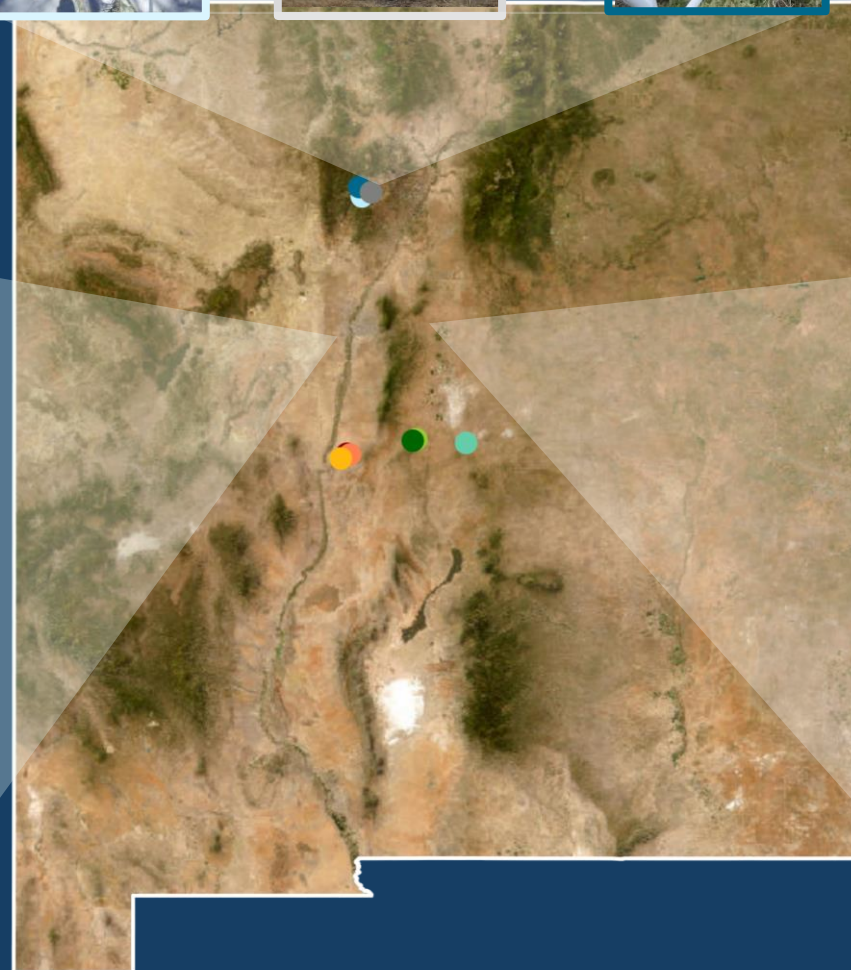
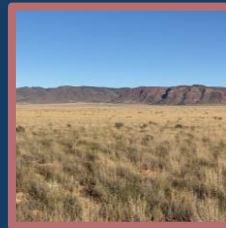
creosote shrubland



unburned grassland



burned grassland



PJ woodland, control



PJ woodland, girdled



Juniper savannah



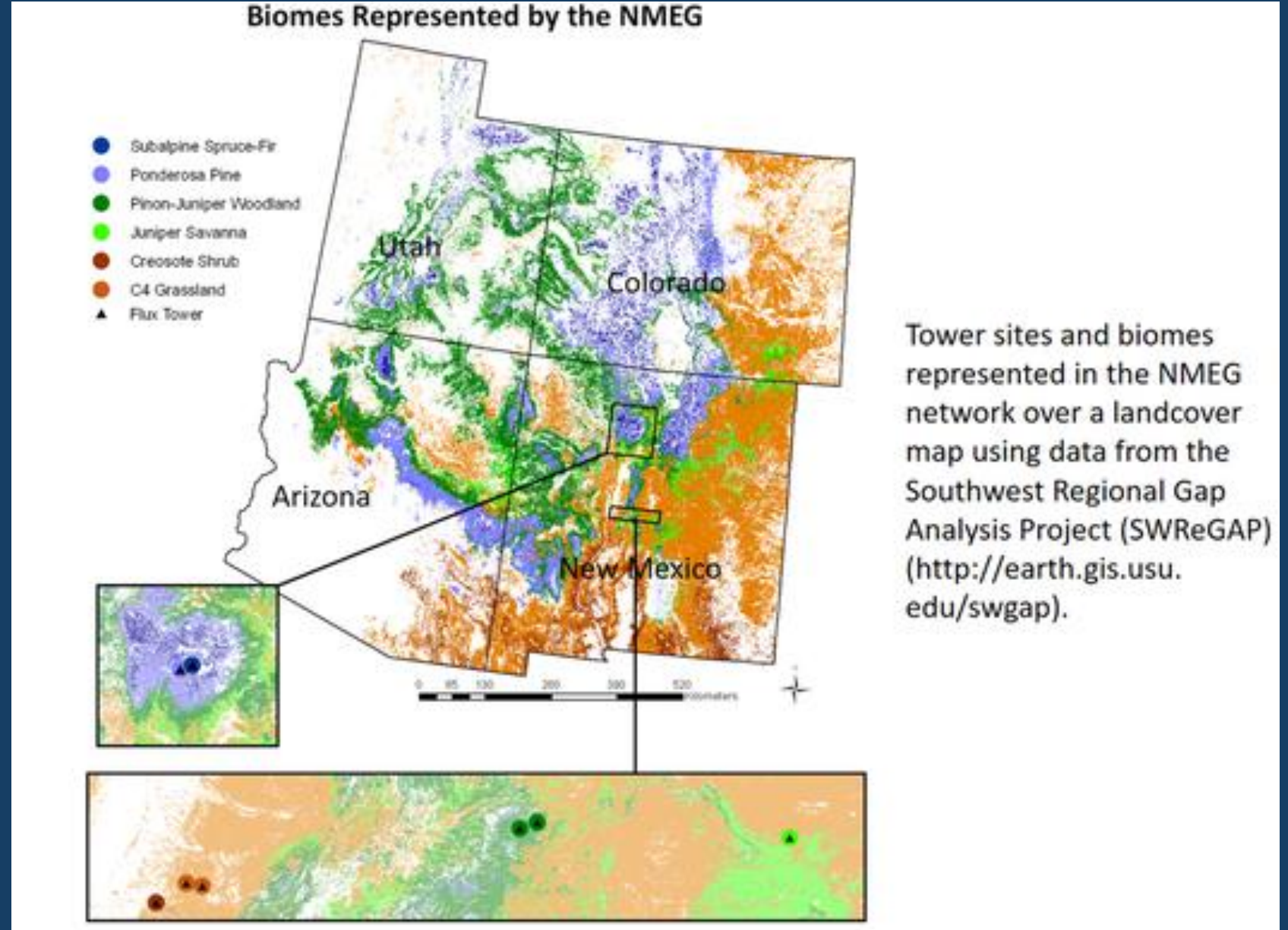
# These biomes span a large range of climatic conditions

|                              | desert<br>grassland                                                                | creosote<br>shrubland                                                               | juniper<br>savannah                                                                 | PJ woodland                                                                         | ponderosa<br>pine forest                                                            | mixed<br>conifer<br>forest                                                          |
|------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|                              |  |  |  |  |  |  |
| elevation (m)                | 1622                                                                               | <b>1593</b>                                                                         | 1931                                                                                | 2196                                                                                | 2500                                                                                | 3000                                                                                |
| MAT (C)                      | 13.67                                                                              | <b>13.72</b>                                                                        | 15.2                                                                                | 10.5                                                                                | 9.8                                                                                 | 6.4                                                                                 |
| MAP (mm)                     | 273                                                                                | <b>273</b>                                                                          | 361                                                                                 | 385                                                                                 | 550                                                                                 | 646                                                                                 |
| % monsoon precip.            | 70                                                                                 | <b>70</b>                                                                           | 53                                                                                  | 45                                                                                  | 45                                                                                  | 36                                                                                  |
| Mean VWC 0-50 cm             | 0.08                                                                               | <b>0.10</b>                                                                         | 0.12                                                                                | 0.11                                                                                | 0.18                                                                                | 0.28                                                                                |
| C.V. SWC 0-50 cm             | 0.31                                                                               | <b>0.24</b>                                                                         | 0.33                                                                                | 0.38                                                                                | 0.28                                                                                | 0.21                                                                                |
| % time in driest<br>quartile | 54                                                                                 | <b>50</b>                                                                           | 58                                                                                  | 54                                                                                  | 33                                                                                  | 11                                                                                  |



And represent a large proportion of the SW United States

- 26.7 million ha in NM
- 80 million ha in SW US





### Data availability

- data:20+ year (22)
- data:15-19 year (31)
- data:10-14 year (55)
- data:5-9 year (176)
- data:1-4 year (163)
- ▲ registered (166)

## AmeriFlux – A network of flux tower sites

617 sites  
447 w/ data

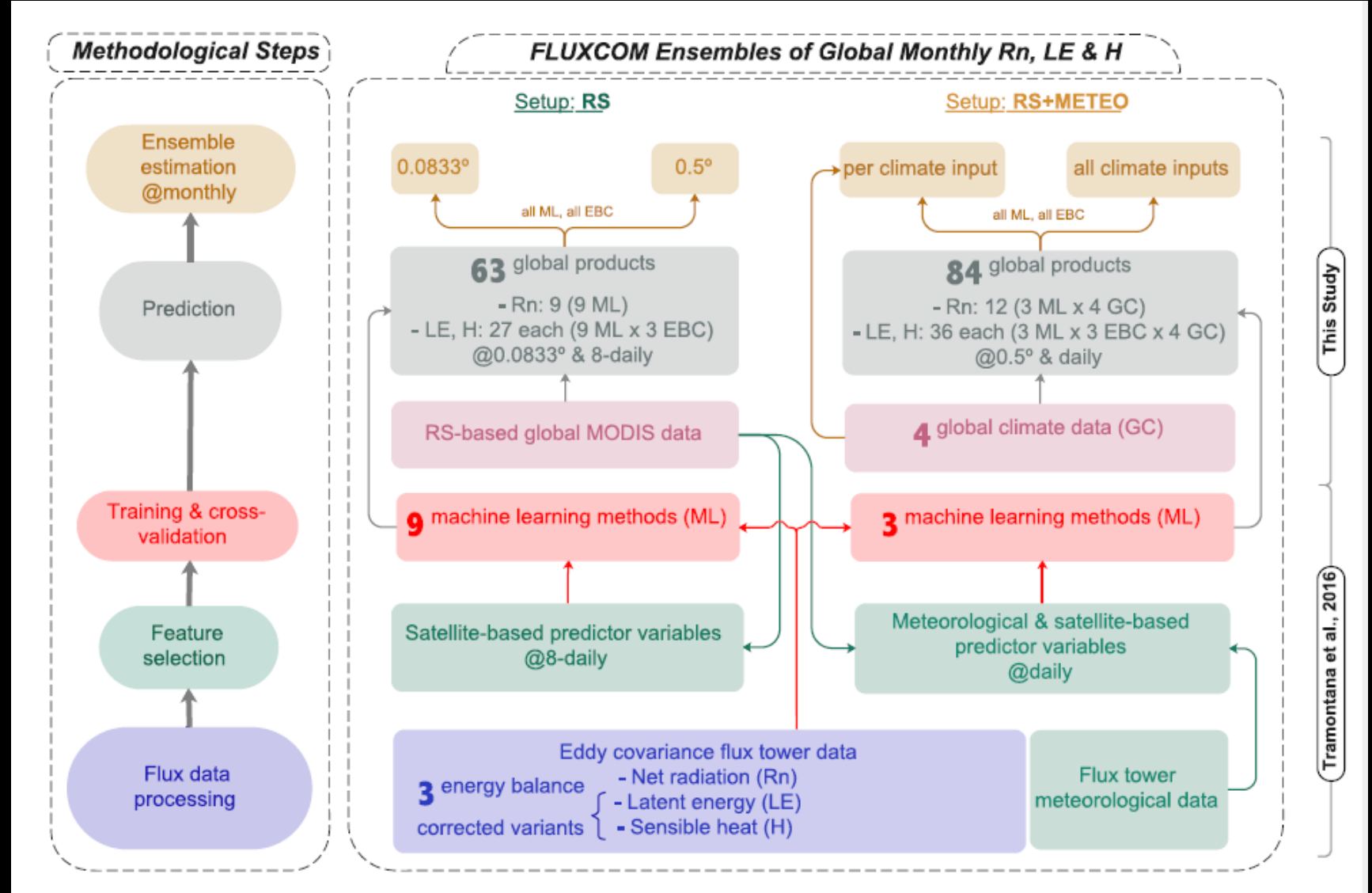


# Fluxnet

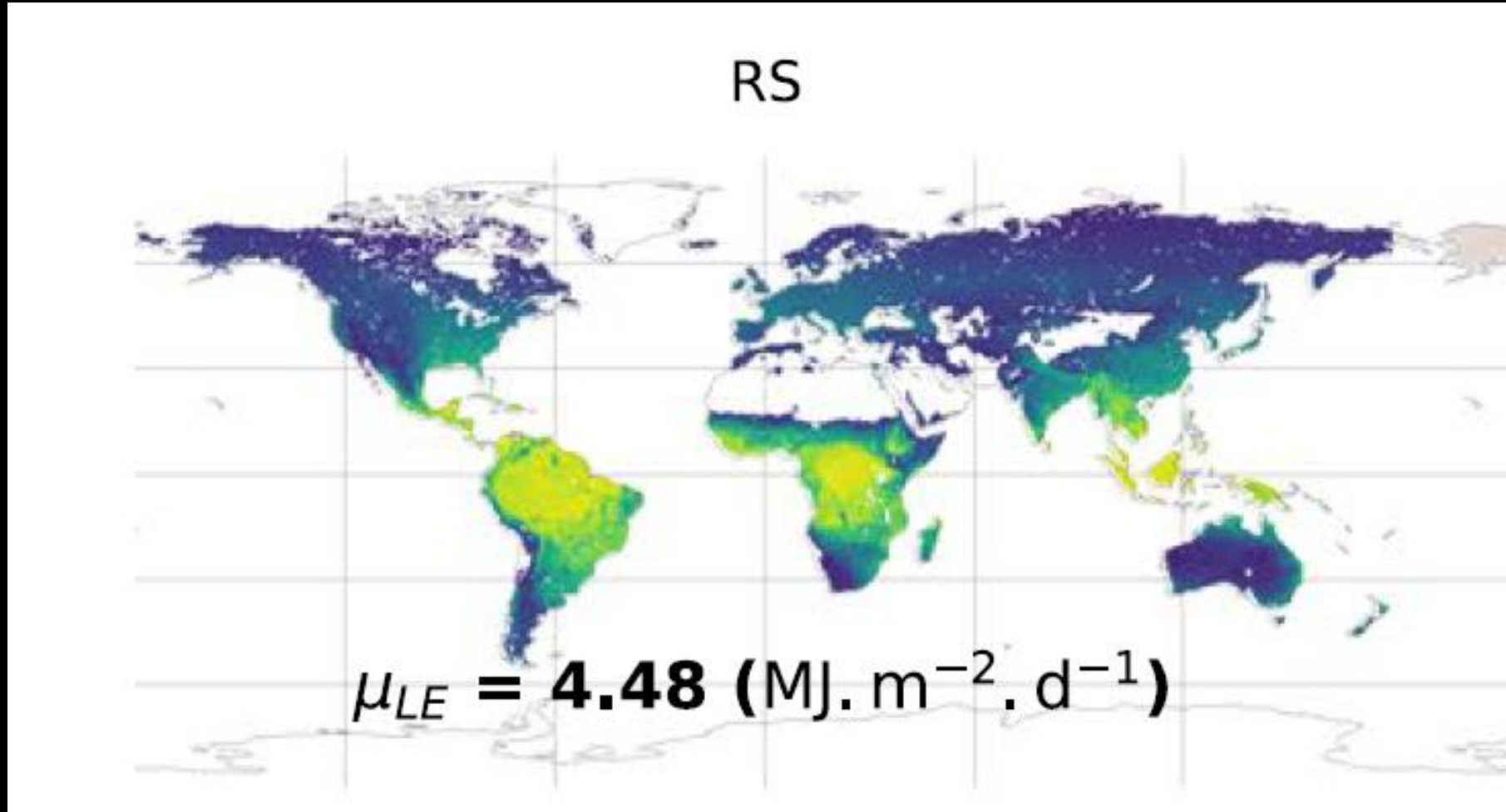


- Combine direct measurements from flux towers, with satellite imagery and models to create U.S. and global scale estimates of ET!

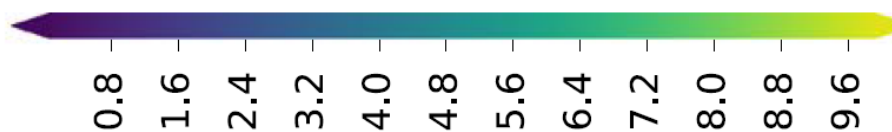
Jung et al. 2019



# Global scale



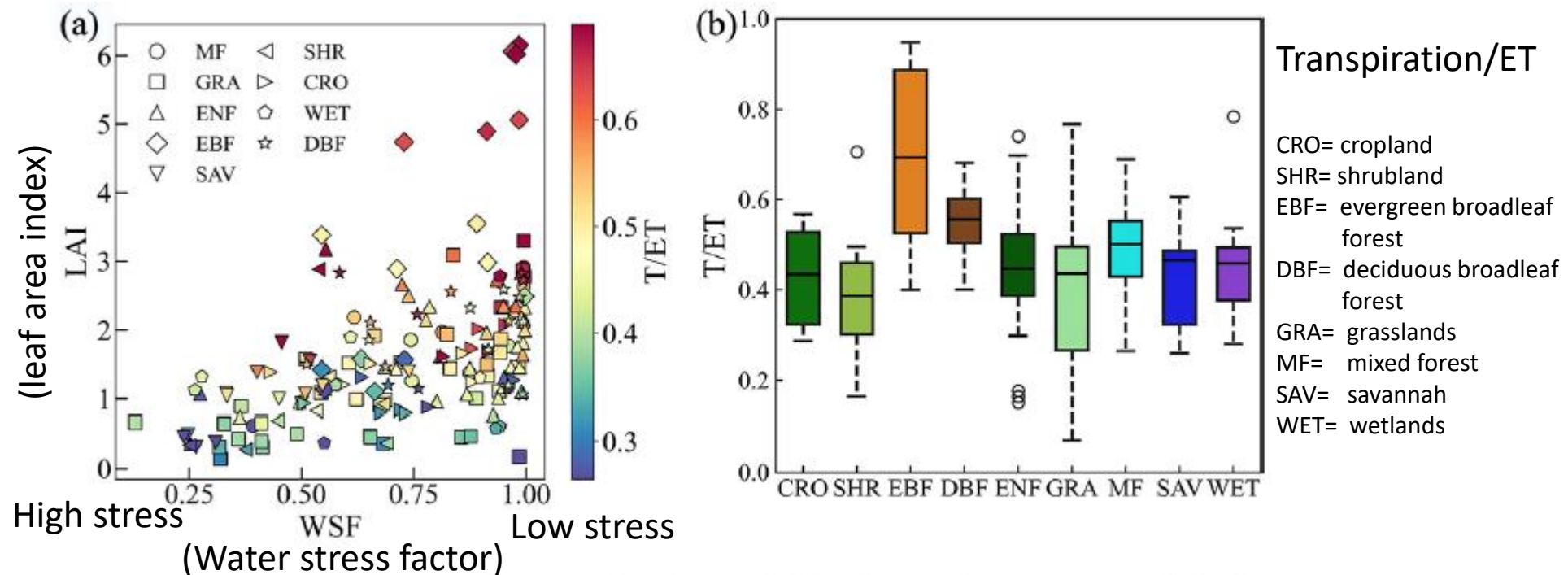
Latent Heat Energy ( $\text{MJ} \cdot \text{m}^{-2} \cdot \text{d}^{-1}$ )



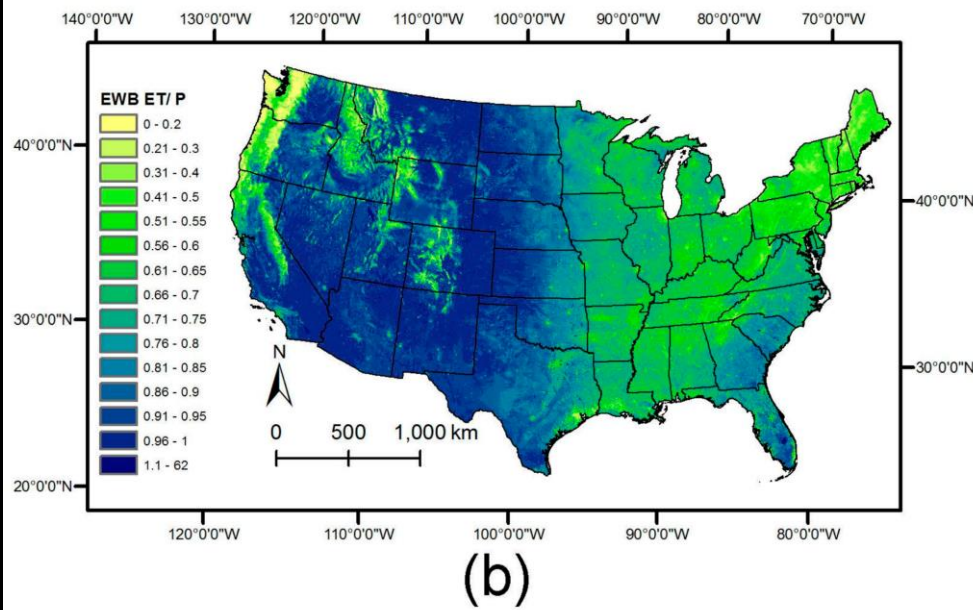
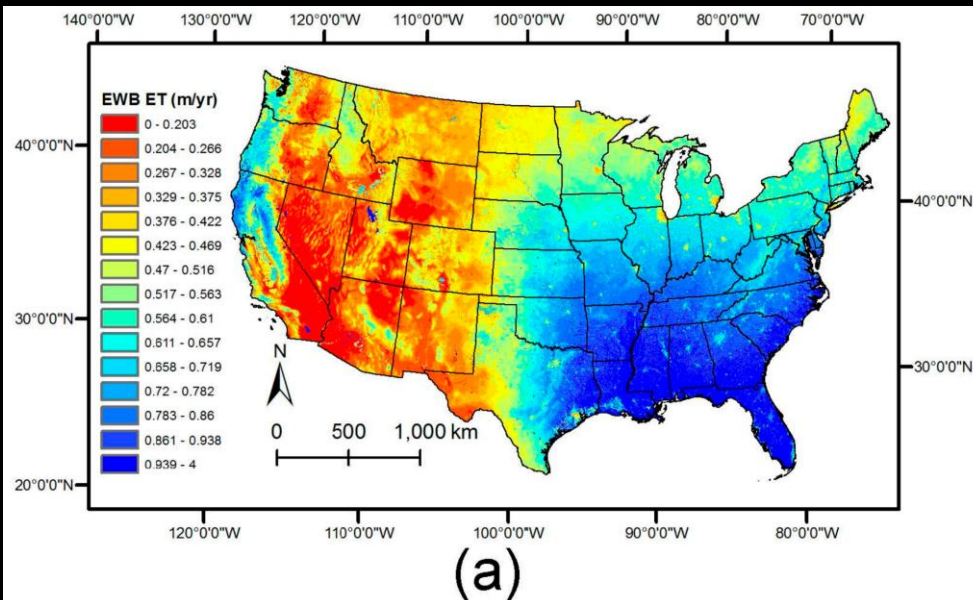
What patterns do you see?  
What explains them?



# What does this tell you about why ET is so high in the tropics?



**Fig. 6** (a) Scatter plot of LAI and WSF with color-coded T/ET ratio for 170 sites with different vegetation types; (b) Boxplot of T/ET ratios for different vegetation types.



- U.S. scale

Where is ET highest?  
Why?

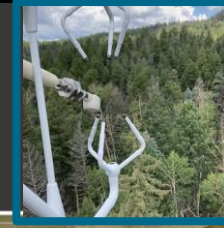
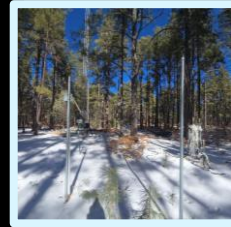
What does ratio of ET/P  
tell you?  
Why is it high in NM?

<https://www.mdpi.com/2072-4292/9/12/1181>

# New Mexico Elevation Gradient

Litvak Lab, UNM  
2007 - present

ponderosa pine      post-fire mixed  
conifer      mixed conifer



What patterns do you expect to see  
in terms of Evapotranspiration?  
Or partitioning between LH and SH flux?

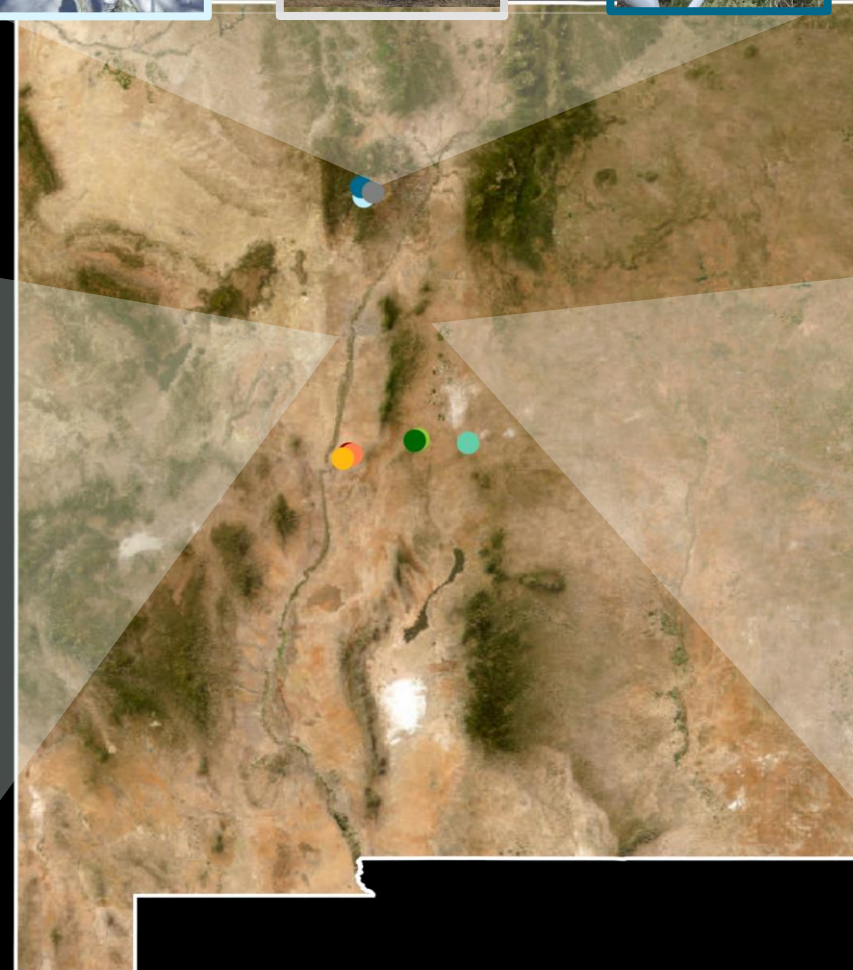
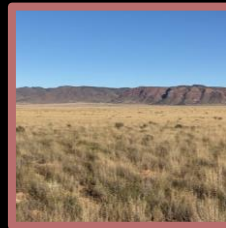
creosote shrubland



unburned grassland



burned grassland



PJ woodland, control



PJ woodland, girdled

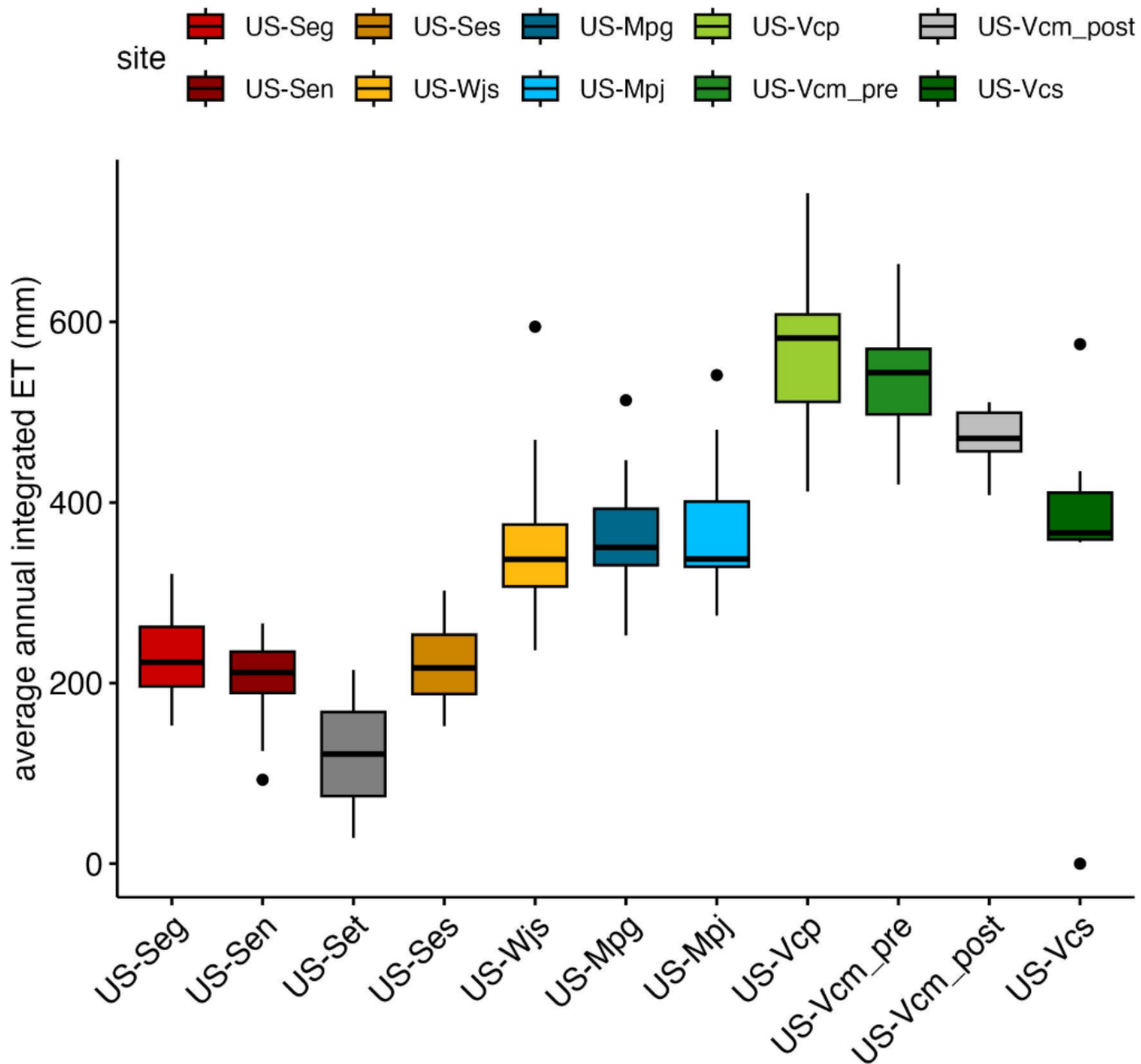


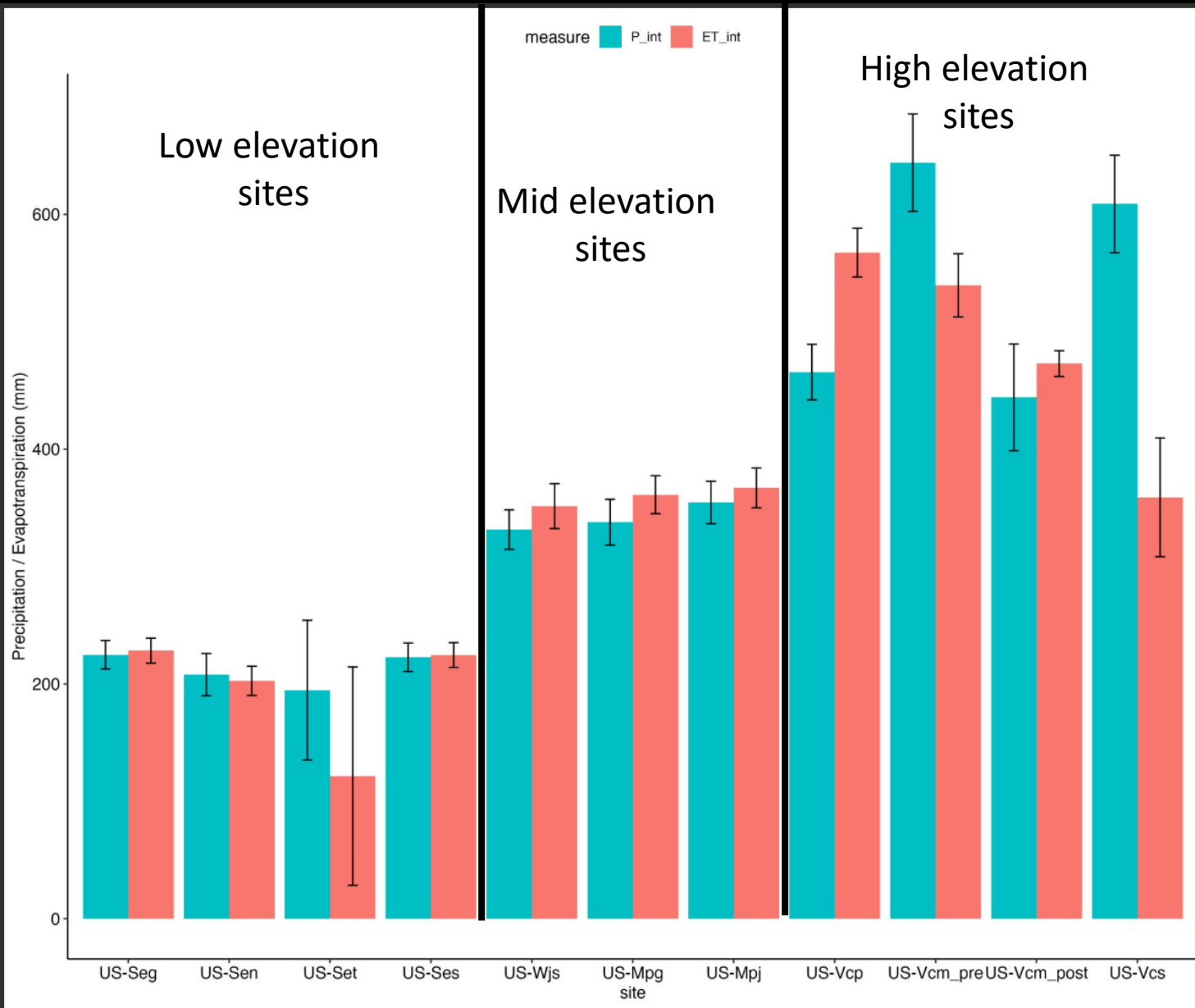
Juniper savannah





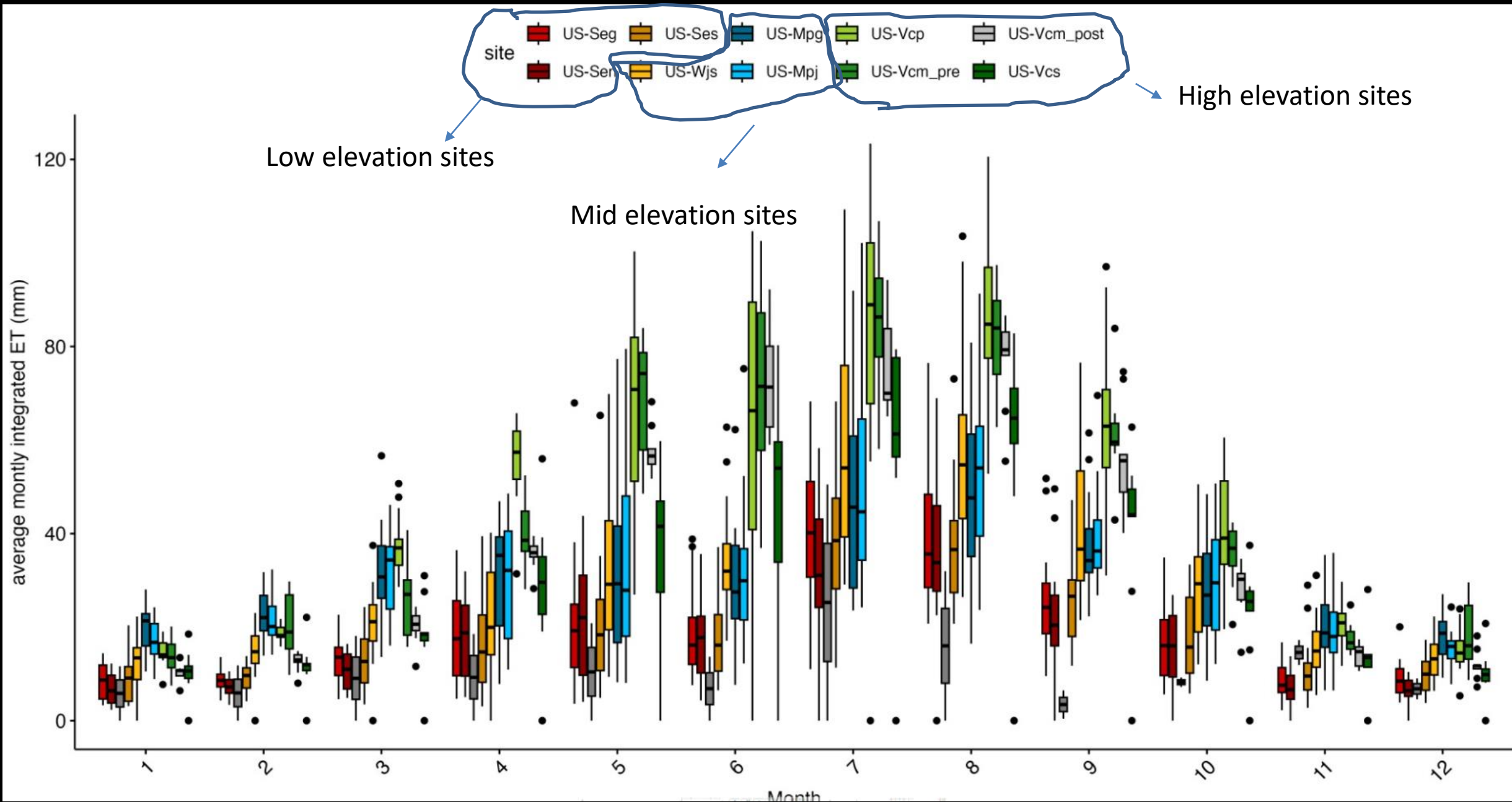
Where is ET  
highest? Does this  
make sense to  
you?



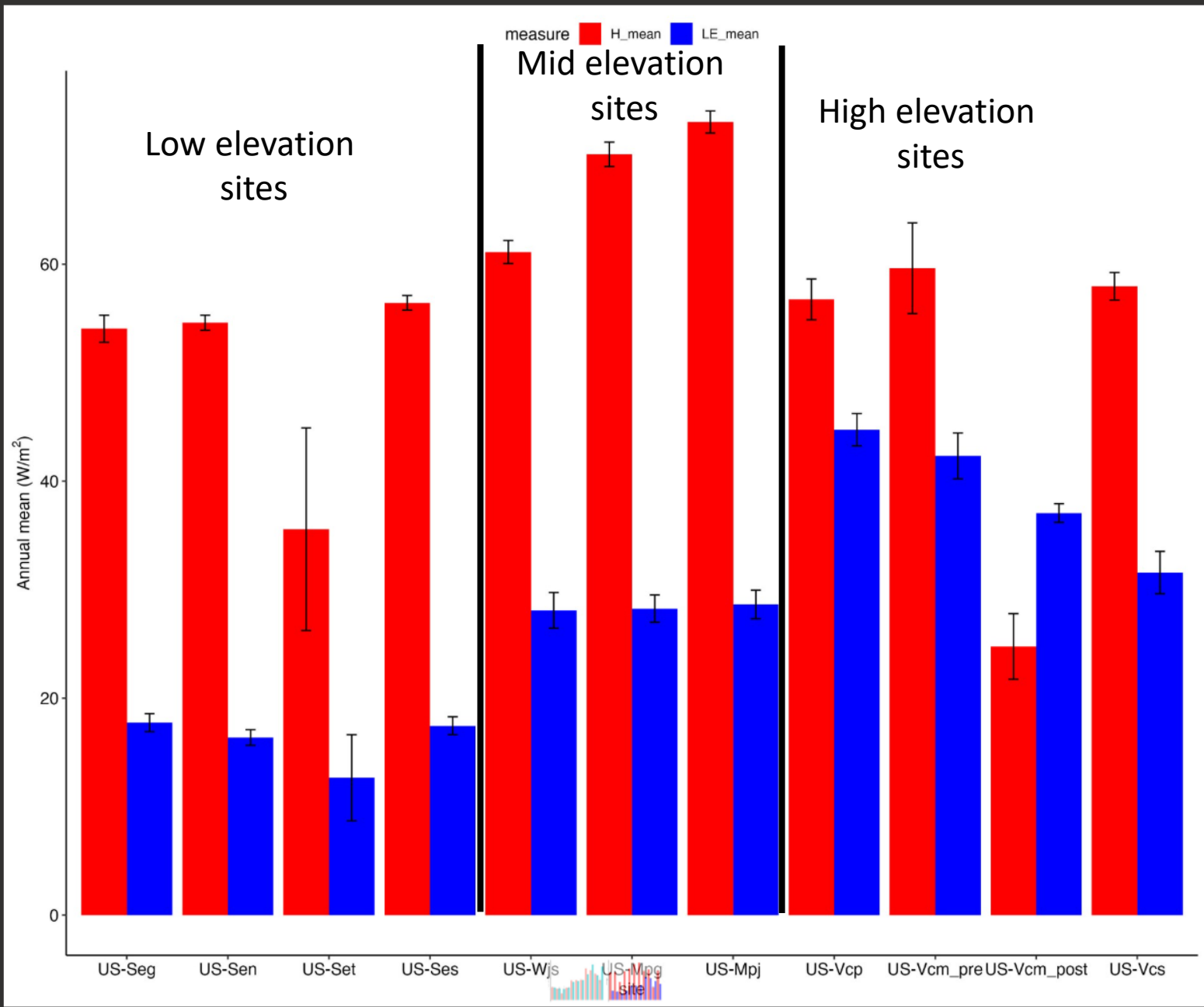


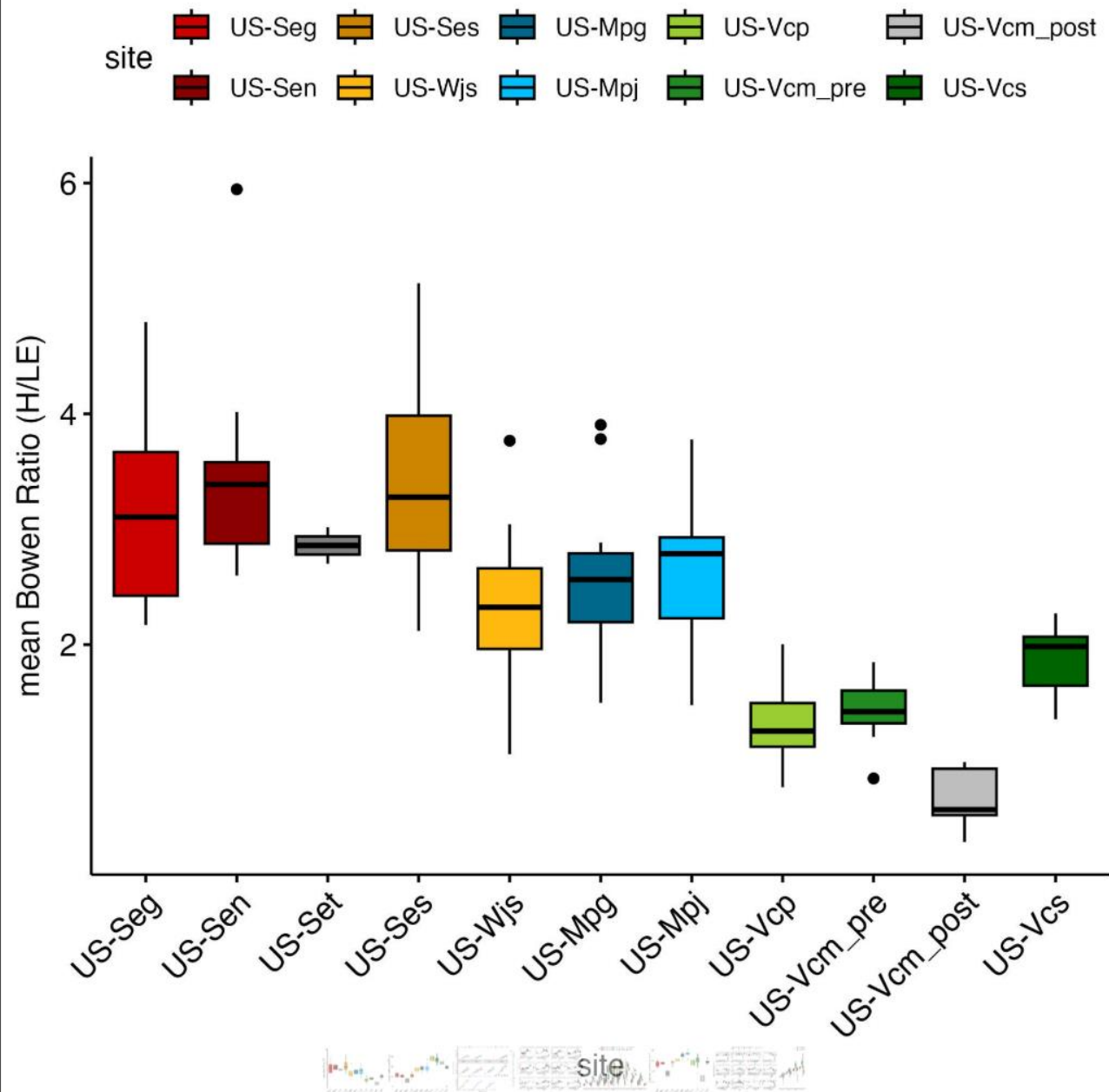
# Patterns across the NMEG

- What do these patterns suggest?
- Can you use these to estimate which biomes are most important for recharge?
- What is going on with Vcp (Ponderosa pine)?









Interannual  
variability in ET  
across the NMEG

What accounts  
for the  
variability?

# Summary

1. Eddy flux requires mass balance of a 'control volume'
2. There are ways to evaluate your measurements, with energy budget closure being the most widely used. There are some problems with it!
3. The instruments themselves have limitations – both in terms of design (line averaging) and theory (WPL).
4. Terrain remains one of the biggest constraints to applying the eddy flux method for ecological studies.

